

Unbalanced Trade: Is Growing Dispersion from Financial or Trade Reforms?*

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Abstract

We study the forces behind the large, coincident increases in international trade imbalances and the overall scale of trade from 1970 to 2019. We show that these two salient features—a rise in net and gross trade—are largely the outcome of reductions in intratemporal trade barriers, rather than substantial declines in intertemporal financial frictions or greater asymmetries in business cycles. Beyond explaining changes in the distribution of net and gross trade, the decline in intratemporal trade frictions is consistent with reduced dispersion across countries in other key macro time series, including the real exchange rate, terms of trade, export-import ratio, relative spending, and relative GDP.

JEL Classifications: E3, F12, F13, F14

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1 Introduction

Global trade has become substantially more unbalanced since the 1970s. Countries now run larger trade deficits and surpluses than they did 50 years ago. As shown in Figure 1, the distribution of trade balances for 183 countries in the 1970s is more tightly centered around zero than in the 2010s. The median absolute deviation from balanced trade nearly doubles, from 5.4 percent to 9.4 percent. While global trade imbalances have risen, their underlying causes remain poorly understood. As a result, they have become politically contentious and have influenced policy decisions.¹ The aim of this paper is to quantify the key forces behind the large increase in global trade imbalances from 1970 to 2019.

Our paper makes several contributions. First, we offer a new way to organize the changes in the distribution of economic activity over time. Specifically, we study how the cross-country dispersion in key macroeconomic time series—output, expenditures, relative prices, and net trade—change with trade. Second, we develop a stochastic multi-country model with rich trade and financial frictions to confront these changes. The model incorporates heterogeneity from the quantitative trade literature, uncertainty and business cycle features from the international macro literature, and both intratemporal and intertemporal frictions that shape borrowing and lending. Third, we emphasize that two trends are central to identifying structural changes in the global economy: the increasing cross-country dispersion in the ratio of net trade to total trade, and the growing synchronization of domestic expenditures across countries. The changes in these two moments are useful to distinguish between trade reforms and financial reforms in the rise in cross-country borrowing and lending.

Fundamentally, three conditions must hold for a country to have unbalanced trade. First, a country must have a reason to run a trade surplus or deficit – arising different economic conditions in the past, present, or future. Second, it must actually trade goods or services. There is no trade between Earth and Mars; hence, trade is balanced between them. Third, countries must trade financial assets, as these assets are needed to finance trade surpluses or deficits. Thus, changes in the distribution of unbalanced trade across countries can reflect changes in these three elements: asymmetric business cycles, trade barriers, or financial

¹For instance, since Donald Trump was first elected U.S. President, U.S. trade policy has changed with a goal of reducing the U.S.’s persistent and large trade deficit.

frictions.

Several observations suggest that declining trade barriers are the primary driver of the rise in trade imbalances. First, country-specific TFP shocks have not become more pronounced over time, implying that asymmetric business cycles are unlikely to be the main driver of global imbalances.² Second, there is a strong correlation between the growth in trade imbalances and gross trade in the data. Figure 2 scatters the dispersion of cross-country net trade—measured by trade balance to GDP ratio (hereafter, TBY)—against the mean of the scale of trade, measured by the gross trade to GDP ratio (hereafter, TRY), for each year from 1970 to 2019. There is a positive relationship summarized by the regression line.³ Third, that declines in international trade barriers can explain rising borrowing and lending is intuitive since a country closed to international trade is also closed to intertemporal trade. As a country removes barriers to trade in goods and services, the impact of business cycle asymmetries on intertemporal trade and the trade balance will be amplified.

We evaluate the relative contribution of trade and financial integration to the rise of global trade imbalances. We focus on these two aspects of global integration as the main forces that could determine larger movements in both TBY and TRY. We begin with the observation that the cross-country distribution of TBY has increased over time. We propose a simple decomposition of the widening imbalances in TBY that suggests most of these changes are attributed to a larger scale of trade rather than the movements in trade balance *as a share of trade*. Indeed, as a share of trade, we find that dispersion in net trade flows has fallen considerably over time.

We then show how changes in the trade balance as a share of trade relate to changes in the distribution of economic activity across countries. That is, we exploit an Armington trade model, the core trade block in most multi-good international trade and macro models, and relate the trade balance to other international macroeconomic variables determining borrowing and lending. We show that over time movements in international relative prices,

²We also find that cross-country dispersion of output and expenditure has slightly decreased, as we discuss in Section 3.

³The positive relationship between dispersion in the net trade and the level of gross trade also holds in the time series for individual countries. In the lower panel of Figure 2, we take each country as a unit of observation, and scatter the standard deviation of the TBY against the average TRY. Here too, we find a positive relationship, albeit with less explanatory power and some important outliers.

relative income, and relative expenditures have also become more muted. In some sense, countries are more synchronized than before and so there is less incentive to run trade imbalances. Even so, trade imbalances have widened, which points to continued integration and allows the changing behavior of prices, income, and expenditures to separate financial from trade integration forces.

To study how trade and financial frictions shape cyclical fluctuations in the trade balance, we develop a general equilibrium model of business cycles, international trade, and integration. Our model extends the real business cycle model of [Backus et al. \(1994\)](#) to include many heterogeneous countries facing uncertain shocks to TFP, financial frictions, and trade barriers. The goods produced in each country are imperfect substitutes and can be traded subject to iceberg shipping costs. Reducing these shipping costs increases trade. We follow [Baxter and Crucini \(1995\)](#) and [Schmitt-Grohé and Uribe \(2003\)](#) by assuming countries can borrow and lend a non-contingent bond at a country-specific interest rate that increases with debt. We discipline this friction with the movements in net trade flows across countries and over time. Financial integration arises from shocks that reduce the elasticity of borrowing rates to debt.⁴

To match the data, our model includes additional frictions. First, we introduce adjustment frictions on import shares that generate the slow responses emphasized in the literature ([Ruhl, 2008](#)). Second, we allow firms to set destination-specific markups that vary with the real exchange rate, generating incomplete pass-through and helping match movements in the terms of trade and net trade flows ([Burstein and Gopinath, 2014](#)). Third, we include shocks to the interest rate gap between countries. [Devereux and Engel \(2002\)](#) and [Itskhoki and Mukhin \(2021\)](#) suggest that these uncovered interest parity shocks are necessary to capture the high volatility of the real exchange rate, while [Mac Mullen and Woo \(2025\)](#) emphasize these shocks must be paired with dynamic trade adjustment and trade cost shocks to capture net trade flows and business cycle transmission.

We interpret the international time series on business cycles and trade through a model with eight countries exposed to global and country-specific shocks to productivity, trade

⁴This modeling approach to international financial frictions is widely employed in the small open economy business cycle literature, for example [Mendoza \(1991\)](#), [Neumeyer and Perri \(2005\)](#), [Uribe and Yue \(2006\)](#), and [Garcia-Cicco et al. \(2010\)](#).

costs, and financial frictions. The global shocks to cross-border trade and financial frictions drive the global changes in trade integration and cross-country borrowing and lending. They also interact with the country-specific shocks to capture the changing nature of cross-border economic activity. Studying these changes in the properties of business cycles requires solving the model with a second order approximation.

The model is calibrated to be consistent with key features of business cycles and the distribution of economic activity across countries and over time. It successfully matches the observed increase in the dispersion of TBY with trade, and the fall in the dispersion in the trade balance as a share of trade, relative prices, and spending with trade. We show that most of these patterns arise from a decline in global trade costs, or trade integration. Holding constant the financial friction to international borrowing changes borrowing patterns very little. However when trade frictions are held constant, we are unable to increase the level of trade nor explain how the cross-country nature of economic activity changes with international trade. Moreover, net trade flows become too dispersed for a given level of trade.

Our analysis offers several new ways to study the international time series on business cycles and integration. On the data side, guided by theory, we propose new decompositions of the changing nature of business cycles related to trade. These decompositions identify the source of the changes in the structure of the global economy. On the model side, we build a large multi-country model and estimate the key parameters. Unlike other work with two countries and uncertainty or many countries under perfect foresight with a limited set of margins, our analysis has both elements. Moreover, we emphasize that high-order perturbation methods are important to quantify the relationship between trade and financial integration. Our model analysis enables an accounting of the empirical relationships we have uncovered on the changing nature of the macroeconomy with trade and financial integration.

Section 2 relates our paper to previous work. In section 3 we document how the distribution of economic activity across countries has changed as the scale of trade increases. In section 4 we develop a stochastic multi-country model. In section 5 we relate the properties of the model to the data. We show how changes in intratemporal and intertemporal trade frictions affect the properties of business cycles. We also demonstrate the key role of some of our modeling assumptions about trade dynamics, pricing, the process on trade shocks, the

asset structure of trade, and solution method. Section 6 concludes.

2 Related Literature

Our paper relates to an extensive literature on the determinants of international capital flows. Early work on cross-border capital flows focused on the high correlation between domestic savings and investment rates, following [Feldstein and Horioka \(1980\)](#). [Tesar \(1991\)](#) shows that the saving-investment puzzle is substantially mitigated when there are barriers to international trade. An expansive literature attributes the high correlation to financial market incompleteness ([Bai and Zhang, 2010](#); [Chang et al., 2013](#)). [Gourinchas and Jeanne \(2013\)](#) also study the dynamics of capital flows from 1980 to 2000.

Our work also relates to the literature on international risk sharing. [Lewis \(1996\)](#) uses a large sample of countries to demonstrate the lack of international risk sharing. [Backus and Smith \(1993\)](#) test international risk sharing with consumption and real exchange rate data. [Heathcote and Perri \(2004\)](#) study the decline in consumption co-movement between the United States and Europe following an increase in cross-border equity flows. [Heathcote and Perri \(2013\)](#) study how cross-border equity flows increase with reductions in trade costs. [Bai and Zhang \(2012\)](#) explains why there is little improvement in international risk sharing among developed and emerging economies after an increase in international debt flows. Our paper considers both trade and financial frictions in a many country general equilibrium model. We use the salient features of cross-country capital flows, relative prices, and trade integration to disentangle the importance of the two frictions. More recent work focuses on small open economy two-good models and pricing frictions in international transmission, but largely abstract from the properties of the trade balance ([McLeay and Tenreyro, 2025](#)).

Recent work attributes the growth in trade imbalances to financial asymmetries across countries. For instance, [Choi et al. \(2008\)](#) and [Kehoe et al. \(2018\)](#) attribute the rise in U.S. borrowing and China's lending to differences in the discount factor across countries. Likewise, [Mendoza et al. \(2009\)](#) attribute these different financial conditions to differences in the country-specific idiosyncratic risk in the presence of market incompleteness, while [Caballero et al. \(2008\)](#) emphasize differences in the ability of countries to produce safe

assets. Most analyses abstract from changes in trade integration over time and often focus on one-good models.⁵

Our paper expands on recent efforts to bridge the gap between international trade and international macroeconomics and finance. Starting with [Obstfeld and Rogoff \(2000\)](#), a series of papers have explored the role of trade barriers in aggregate fluctuations and international capital flows, ([Alessandria and Choi, 2021](#), [Barattieri, 2014](#), [Eaton et al., 2016](#), [Fitzgerald, 2012](#), and [Mac Mullen and Woo, 2025](#)). Most closely related is [Alessandria and Choi \(2021\)](#) who study the role of trade integration in explaining the growing trade deficits of the U.S. over time in a two country model of the U.S. and the rest of the world.⁶ Here we consider the effects of integration and borrowing and lending for a much broader set of countries. Our model also includes capital and a richer set of financial shocks. Our paper departs from the large multi-country models, that since [Dekle et al. \(2007\)](#) largely treat trade balances as exogenous shocks. In this spirit, several recent papers also study the effects of trade integration on cross-border financial flows in many country international trade models (see [Reyes-Heroles, 2016](#) and [Sposi, 2022](#)).⁷ Unlike these papers, which focus on a perfect foresight economy, we explicitly allow for uncertainty about trade policy and aggregate shocks. These added features are necessary to capture the nature of business cycles and the impact of uncertainty on international risk sharing and capital flows. Our model also brings in several important frictions related to trade adjustment and pricing that are common in the international macro literature and matter for our quantification.

Building on the work of [Kose and Yi \(2006\)](#), several papers have studied how business cycles have changed with trade integration ([Caselli et al., 2019](#); [Bonadio et al., 2021](#); [Miyamoto and Nguyen, 2024](#); [Avila-Montealegre and Mix, 2024](#)). Our paper differs in its focus on the changes in the distribution of trade imbalances⁸ and the distribution of economic activity

⁵Recent prominent examples of analyses of current account dynamics in one-good models include [Ohanian et al. \(2018\)](#), [Ohanian et al. \(2023\)](#), [Atkeson et al. \(2025\)](#), and [Mendoza and Quadriini \(2023\)](#).

⁶[Kuipers et al. \(2024\)](#) focus on the relative importance of financial and trade frictions in the changes of the US trade deficit over time. [Fitzgerald \(2025\)](#) studies how international trade improves risk sharing in an endowment economy.

⁷[Cuñat and Zymek \(2024\)](#) also study imbalances but focus on the source of bilateral imbalances rather than the growth in aggregate imbalances. [Dix-Carneiro et al. \(2023\)](#) study the dynamics of trade integration in a model of labor adjustment and unbalanced trade. [Kleinman et al. \(2023\)](#) study a multi-country model of capital and trade flows but with different capital market frictions and focus on transition dynamics.

⁸Much of this work abstracts from trade imbalances, see [Caselli et al. \(2019\)](#); [Bonadio et al. \(2021\)](#).

across countries as well as in our modeling of trade as a stochastic process and explicitly accounting for the transition to a more open economy.

3 Empirics

To uncover the forces driving cross-country borrowing and lending, we examine how net and gross trade flows evolve over time and across countries. Our empirical strategy is guided by the constant elasticity of substitution (CES) demand system of the Armington trade model, which links net trade flows to macroeconomic asymmetries—specifically, differences in relative prices, relative expenditures, and the scale of trade. Using this framework, we document a key empirical pattern: although net trade flows as a share of GDP have become more dispersed, this dispersion is largely driven by rising gross trade volumes rather than increased imbalances relative to trade. In parallel, macroeconomic asymmetries—such as differences in business cycle conditions—have diminished over time. These stylized facts serve as empirical targets for our model and motivate the decomposition and estimation strategy that follows.

3.1 Organizing Structure

Following [Levchenko et al. \(2010\)](#) and [Alessandria and Choi \(2021\)](#), we organize our empirical work around a constant elasticity of substitution (CES) demand system. This framework assumes that net trade flows are tightly linked to gross trade flows, relative prices, relative expenditures, and trade wedges—defined as deviations from theoretically predicted net trade flows due to changes in trade costs and tastes.

Our decomposition proceeds in two steps. First, we mechanically split the trade balance to GDP ratio (hereafter, TBY) into two terms:

$$\underbrace{\frac{X - M}{Y}}_{\text{TBY}} = \underbrace{\frac{X + M}{Y}}_{\text{TRY}} \times \frac{X - M}{X + M} \quad (1)$$

where X is home exports to the rest of world, M is home imports from the rest of the world,

and Y is GDP.⁹ The first term is the trade share of GDP (hereafter, TRY), which captures the overall scale of trade in the economy. This decomposition is particularly useful because, in nearly all models, TRY is primarily determined by trade barriers and country size. The second term is the trade balance as a share of gross trade. Importantly, it can be closely approximated by one-half of the log of the ratio of exports to import:

$$\frac{X - M}{X + M} \approx 0.5 \ln \frac{X}{M}.$$

In the second step, we exploit this approximation and further decompose the export–import ratio, $\ln(X/M)$. To do so, we rely on the Armington model, the standard trade block in most multi-good international macroeconomic frameworks. With imperfectly substitutable home and foreign goods and a CES demand system, exports and imports are given by

$$X = \omega^* \left(\frac{p\xi^*}{P^*} \right)^{-\gamma} D^*, \quad M = \omega \left(\frac{p^*\xi}{P} \right)^{-\gamma} D,$$

where γ is the elasticity of substitution between home and foreign goods (the Armington elasticity), ω denotes the taste or bias for imports, ξ is an ad valorem trade cost or tariff, p is the price of the differentiated good before any trade cost or tariff, P is the price level in domestic, D is domestic spending on tradables, and an asterisk refers to the foreign analogues to a home variable. Notice we allow the cost of exporting to differ from importing, i.e. $\xi \neq \xi^*$. Defining the real exchange rate, $rer = \ln P^*/P$, terms of trade, $tot = \ln p/p^*$, trade wedge, $\tau = \ln(\omega\xi^{-\gamma})$, and expenditures, $d = \ln D$, we can rewrite the export-import ratio as

$$\ln \frac{X}{M} = (\tau^* - \tau) - \gamma(tot - rer) + (d^* - d). \quad (2)$$

Hence, the export-import ratio is determined by cross-country differences in trade wedges, international relative prices, differences in expenditures, and the Armington elasticity. Note that Equation 2 holds regardless of the assumptions on the structure of asset or goods markets, even though these assumptions could influence prices and expenditures. Importantly,

⁹We focus on the decomposition in real trade flows, but this is easily extended to nominal terms with a slight adjustment to the trade elasticity. As our analysis targets the behavior of relative prices and relative spending, matching real or nominal terms will yield similar decompositions.

most terms have clear empirical counterparts.

3.2 Data

We study the properties of these determinants of the trade balance in a sample of 36 countries that includes both developed and emerging countries. We consider countries in the Penn World Table 10.0 for which we have data since 1970 and are covered in the broad basket of the Bank for International Settlements (BIS) Effective Exchange Rates. In Appendix B, we derive similar results for different samples of countries.

Following Equation 1, TBY can be expressed as the product of trade openness, TRY, and the trade balance as a share of trade, $\ln(X/M)$. The solid blue line in Figure 3 plots the dispersion in TBY, measured by the interquartile range across countries in each year.¹⁰ The red dashed line shows that the median level of trade, TRY, rises over time. At the start of the sample in 1970, trade was about 26 percent of GDP. By 2019, trade had almost doubled to 51 percent of GDP. Trade grows persistently until the Great Recession, peaking at nearly 60 percent of GDP, but has since fallen. TBY dispersion rises alongside TRY during periods of expanding global trade, suggesting that as countries become more open, cross-country differences in trade balances also widen. The positive correlation between the two series implies a strong relationship between trade openness and trade imbalances.

To isolate the role of trade in the widening trade imbalances, we construct a counterfactual measure of dispersion that holds each country's TRY constant at its level in 1971.¹¹ With this alternative measure, the positive relationship between dispersion in the trade balance and trade disappears. In fact, this measure of dispersion now falls over time. This suggests that the growth in the dispersion of the trade balance may be attributed to trade integration amplifying the movements in the trade balance to trade ratio.

Taking a cross-sectional perspective, the upper panel of Figure 2 illustrates this positive

¹⁰Alternative measures of dispersion, such as the standard deviation or the 10–90 percentile range, yield similar patterns.

¹¹This is a relatively clean decomposition that highlights the relative importance of trade and the trade balance as a share of trade. In Appendix C we provide a complete variance decomposition that introduces additional terms and decomposes the growth of these terms. Importantly, the combined contribution of the mean of trade multiplied by the standard deviation of the trade balance to trade have remained the main determinants of dispersion in the trade balance to GDP.

relationship between TBY dispersion and mean TRY in a scatter plot.¹² That is, the increase in trade is associated with an increase in dispersion in borrowing and lending. The elasticity of dispersion to trade is about 11 percent. The lower panel repeats this analysis using countries, rather than time periods, as the unit of observation. Each point represents a country’s trade balance volatility and mean trade share over the sample period. The positive slope suggests that countries that are more open to trade tend to experience greater fluctuations in their trade balances. This confirms the robustness of the positive relationship between trade openness and trade balance volatility. To compare, Figure 4 plots the annual cross-country dispersion of the export–import ratio. Dispersion declines as mean TRY rises, with an elasticity of around 38 percent. The pattern also holds even when using the country as a unit of observation, as shown in the lower panel. Countries with a larger scale of trade tend to exhibit less dispersed net trade as a share of GDP, whereas dispersion decreases once the scale of trade is held constant.

These two graphs imply that the growth in trade, rather than the increase in the trade balance over trade, accounts for the rising dispersion in the trade balance. In other words, the widening cross-country variation in trade balances is largely driven by the expansion of trade, which amplifies the scale of international borrowing-lending. When we abstract from this growth in trade and hold the trade share constant, the dispersion in trade balances no longer increases and, in fact, tends to decline.

Taken together, counterfactuals from the time series in Figure 3 and the cross-sectional evidence in Figures 3 and 4 indicate that growth in TRY is the primary driver of the observed rise in TBY dispersion. This pattern is consistent across the data and is not driven by a few large countries.¹³

We next use our theoretical decomposition as in Equation 2 to uncover the source of the decline in the cross-country dispersion in the export–import ratio with trade. Specifically, we construct measures of the cross-country dispersion of the terms of trade, real exchange rate,

¹²Using median, instead of mean, to measure overall scale of TRY yields very similar results.

¹³The pattern is not driven by one or two countries such as the US or China. A similar pattern is observed when using alternative measures, such as the interquartile range and/or the median, both of which are largely unaffected by outliers. Moreover, the relationship remains similar when using countries as the unit of observation, and excluding one or two countries does not materially alter the result. Appendix B provides results based on alternative country samples.

and domestic expenditures, which is the sum of consumption, investment, and government spending. For dispersion in relative expenditures, for each country we construct a measure of foreign expenditures using the country’s BIS trade weights. Also, for relative prices, we focus on the annual changes rather than the levels.¹⁴ Figure 5 presents the scatter plots of these measures of dispersion against the mean trade share in each year. In all panels of Figure 5, we observe a negative relationship: as the trade share of GDP increases over time, dispersion in relative prices and relative spending falls.

It is worth noting that the growing dispersion in the trade balance, TBY, does not appear to be attributed to a change in the underlying productivity shocks. Figure 6 shows that the cross-country dispersion of TFP, measured with its interquartile range, has not increased during the last five decades. Aggregate output also becomes less dispersed across countries over time, as shown in the lower right panel of Figure 5.

On the other hand, Equation 2 shows that the importance of relative prices, expenditures, and trade costs depends on the Armington elasticity. Our decomposition of net trade flows into observables allows us to estimate the Armington elasticity by treating the trade wedge as a residual, disciplining the relative importance of relative prices and relative expenditures in net trade flows. We consider three types of regressions: in levels, in first differences, and with an error correction term, to allow for different short-run and long-run trade adjustments. For each of the three types of regressions, we consider two cases, one that constrains the coefficient of short-run relative spending to be one as theory suggests, and the other where the coefficient is estimated. Using our panel of 36 countries during the period of 1970-2019, Table 1 reports the results.

The regression in levels does not fit the data very well. When estimated in the first differences, with and without the error correction term, the Armington elasticity is significant, around 0.23, regardless of how we treat the coefficient on expenditures. To distinguish short-run and long-run effects of relative prices on the export-import ratio, we consider the error correction model, which has a long tradition in trade elasticity estimation (see [Hooper and Johnson, 2000](#)). Columns (5) and (6) show that the long-run elasticity is higher than in the short run, closer to 3.3. The gap between the short-run and long-run Armington elasticity

¹⁴We focus on changes to better capture the cyclical movements in relative prices.

suggests we will need a model with a time-varying Armington elasticity lest we attribute movements in net trade to shocks.

4 Model

We now develop a multi-country variation of the canonical international business cycle model of [Backus et al. \(1994\)](#) with global and country-specific shocks to trade and financial frictions. In each country, there is a non-tradable final good used for consumption and investment, made by combining a different mix of imperfectly substitutable intermediates from all countries. Country-specific intermediates are produced with local capital and labor. There are country-specific shocks to the productivity of producing these intermediates. Trading these intermediates across countries is subject to a stochastic bilateral iceberg trade cost. We also incorporate adjustment costs in the use of imported inputs to produce the final good, and intermediate producers set a destination-specific price (pricing-to-market) as these help to explain the dynamic pattern between relative prices and relative trade flows.¹⁵

As our aim is to allow for changes in trade and financial barriers over time, we assume financial markets are incomplete.¹⁶ We take the simplest model of financial market incompleteness that allows for cross-country borrowing and lending. As in [Baxter and Crucini \(1995\)](#), [Heathcote and Perri \(2002\)](#), and [Schmitt-Grohé and Uribe \(2003\)](#), the consumers can trade a non-contingent bond denominated in units of the final good of country 1. Beyond being non-contingent, the interest rate is debt-elastic so that taking on more debt increases the borrowing rate. We allow for shocks to the debt elasticity parameter to capture changes in financial integration. Following [Devereux and Engel \(2002\)](#), we introduce country-specific uncovered interest parity (UIP) shocks to the international borrowing rate that create an additional wedge between the returns to saving across countries. Recent quantitative studies assign a large role to these shocks in international transmission.

¹⁵See [Drozd and Nosal \(2012\)](#), [Burstein and Gopinath \(2014\)](#), [Rabanal and Rubio-Ramirez \(2015\)](#), and [Alessandria and Choi \(2021\)](#).

¹⁶With complete financial markets, we get similar results between trade integration and the dispersion of the trade balance, but with a higher level of overall trade imbalances.

Consumers Consumers in country n choose consumption, leisure, investment, and bonds to maximize welfare

$$\max E_0 \sum_{t=0}^{\infty} \beta^t u(c_{nt}, \bar{h}_n - h_{nt})$$

subject to a sequence of budget constraints

$$c_{nt} + i_{nt} + e_{nt} q_{nt} b_{nt+1} = w_{nt} h_{nt} + r_{nt}^k k_{nt} + e_{nt} b_{nt} + \Pi_{nt}$$

where $u(c_{nt}, \bar{h}_n - h_{nt}) = \frac{[c_{nt}^\mu (\bar{h}_n - h_{nt})^{1-\mu}]^{1-\sigma}}{1-\sigma}$, q_{nt} is the country-specific discount rate of a non-contingent bond denominated in units of country 1 consumption, $e_{nt} \equiv P_{1t}/P_{nt}$ is real exchange rate defined as the final good price relative to country 1, and Π_{nt} is the dividend payments from domestic firms. The evolution of the capital stock is given by

$$k_{nt+1} = (1 - \delta_k) k_{nt} + i_t - \frac{\psi}{2} \left(\frac{k_{nt+1}}{k_{nt}} - 1 \right)^2 k_{nt},$$

where we include a quadratic capital adjustment cost. We assume the country faces an interest rate that depends on the endogenous world interest rate r_t , the debt of the country relative to its size, $\frac{\bar{b}_n}{Y_{nt}}$, and a country-specific interest rate shock ϕ_{nt} ,

$$1/q_{nt} = r_t + F_t \left(e^{-\left(\frac{b_{nt}}{Y_{nt}} - \frac{\bar{b}_n}{Y_n}\right)} - 1 \right) + (e^{\phi_{nt}-1} - 1)$$

where F_t governs the interest rate elasticity to debt. Note that including the share of debt to output makes debt more elastic for less productive countries. Let λ_{nt} be the marginal utility of consumption. We can define the consumers' stochastic discount factor Λ_{nt} as $\Lambda_{nt} = \frac{\lambda_{nt}/P_{nt}}{\lambda_{0t}/P_{n0}}$.

Final good producers Final good producers are competitive and combine all home and foreign intermediates with a CES aggregator. To allow for the short-run response of trade to changes in relative prices to differ from the long-run one, we follow [Ravn et al. \(2010\)](#) and assume deep habits in import demand, which has the feature of habit formation in the level

of individual goods. Specifically, the final good production D_{nt} in country n is given by

$$D_{nt} = \left\{ \sum_{m=1}^N \omega_{nm}^{\frac{1}{\gamma}} \left[\frac{a_{nmt}}{(a_{nm,t-1}/D_{nt-1})^{\frac{-\delta}{\gamma(1-\delta)-1}}} \right]^{\frac{\gamma(1-\delta)-1}{\gamma(1-\delta)}} \right\}^{\frac{\gamma(1-\delta)}{\gamma(1-\delta)-1}}$$

where a_{nmt} is the intermediate good produced in country m at time t , γ is the long-run elasticity, and the parameter δ determines the degree of deep habit. When $\delta = 0$ we recover the standard CES function.

Final goods producers at country n incur an iceberg trade costs, ξ_{nmt} , when importing goods from country m . Taking as given the aggregate prices P_{nt} and the intermediate goods prices $\{p_{nmt}\}$, a final good producer chooses inputs $\{a_{nmt}\}$ to solve the following problem,

$$\max \quad E_0 \sum_t \Lambda_{nt} \left[P_{nt} D_{nt} - \sum_{m=1}^N p_{nmt} \xi_{nmt} a_{nmt} \right].$$

The implied demand function is given by

$$\ln \frac{a_{nmt}}{D_{nt}} = (1 - \delta) \left[\ln \omega_{nm} - \gamma \ln \frac{p_{nmt} \xi_{nmt}}{P_{nt}} \right] + \delta \ln \frac{a_{nm,t-1}}{D_{nt-1}}. \quad (3)$$

Note that we assume that each final goods producer views $a_{nm,t-1}$ as an economy-wide value and does not internalize the impact of its current choice of $a_{nm,t}$ on future D_{nt+1} .

Consider a case with two countries $\{H, F\}$ and let variables with an asterisk denote foreign variables. Under the demand function (3), the export-import ratio equals,

$$\ln \frac{X_t}{M_t} = (1 - \delta) \ln \frac{\omega_t^* \xi_t^{*- \gamma}}{\omega_t \xi_t^{- \gamma}} + (1 - \delta) \gamma \ln \left(\frac{p_{Ft} P_t^*}{p_{Ht}^* P_t} \right) + \delta \ln \frac{X_{t-1}}{M_{t-1}} - \delta \frac{D_{t-1}^*}{D_{t-1}} + \ln \frac{D_t^*}{D_t},$$

This condition closely relates to the ECM specification in the empirical section.

Intermediate good producers An intermediate goods producer uses domestic labor, h_{nt} , and capital, k_{nt} , to produce a differentiated product with a Cobb-Douglas production function $y_{nt} = z_{nt} k_{nt}^\alpha h_{nt}^{1-\alpha}$ where z_{nt} is the productivity. Taking as given the prices

$(p_{nt}, P_{nt}, w_{nt}, r_{nk}^k)$, a producer solves the following problem,

$$\max_{h_{nt}, k_{nt}} \frac{p_{nt}}{P_{nt}} y_{nt} - w_{nt} h_{nt} - r_{nt}^k k_{nt}.$$

In the data, the real exchange rate is more volatile than the terms of trade and this increased volatility has been attributed to pricing to market (PTM). We take a simple approach to modeling PTM and assume that firms charge a markup over marginal cost that is a function of local market conditions proxied by the real exchange rate.¹⁷ This pricing structure implies that when the dollar appreciates all firms selling in the U.S. charge high markups while all firms selling outside the U.S. would reduce their markup.¹⁸ Specifically, the price of country n producer selling to country m , p_{mnt} is given by

$$\frac{p_{mnt}}{P_{mt}} = \frac{\gamma \left(\frac{P_{nt}/P_{mt}}{P_n/P_m} \right)^\theta}{\gamma \left(\frac{P_{nt}/P_{mt}}{P_n/P_m} \right)^\theta - 1} \cdot \frac{1}{z_{nt}} \left(\frac{1}{\alpha} \right)^\alpha \left(\frac{1}{1-\alpha} \right)^{1-\alpha} r_{nt}^{k\alpha} w_{nt}^{1-\alpha},$$

where θ controls the degree of pricing-to-market. When $\theta = 0$, there is a constant markup.

Equilibrium In equilibrium, consumers and firms in each country take as given prices and optimize their decisions. The following market clearing conditions hold: $D_{nt} = c_{nt} + i_{nt}$, $y_{nt} = \sum_m \xi_{mn,t} a_{mn,t}$, and $\sum_n b_{nt} = 0$.

5 Quantitative Analysis

Here we describe our parameterization of the model and how we choose parameters to match data moments on trade integration, trade balance dynamics, international relative prices, and business cycles. We then explore the workings of our model by turning off different sources of international integration. These experiments point to changes in international trade barriers mattering most for the growth in unbalanced trade across countries over time.

¹⁷Alessandria and Kaboski (2011) and Drozd and Nosal (2012) show that a model with consumer search frictions yields price setting with these features. This formulation is used by Alessandria and Choi (2021), Mac Mullen and Woo (2025) and Kekre and Lenel (2024)

¹⁸This formulation can be justified with a nested CES framework in which country specific varieties are differentiated and the markup depends on the real exchange rate in the way described above.

The model economy is calibrated to 8 asymmetric countries. This economy captures the key moments of our sample of 36 countries described in section 3. Although this is a sparse representation of the world economy, adding more countries does not alter the relationship between the cross-country dispersion in borrowing and lending and the mean trade share (see Section 5.4). The main challenge to including more countries is computational, as a second order approximation of the model is required to capture the changing nature of business cycles with trade and financial integration. We show that solving the model with a first order approximation understates the relationships highlighted.

5.1 Parameterization

We begin by describing the stochastic processes for shocks, which include bilateral trade costs, international financial frictions, and productivity. For most shocks, we allow both common and country-specific components.

We assume the iceberg trade cost, ξ_{mnt} , between any pair of countries n and m has two components: a common world trade shock, ξ_{ct} , and a differential bilateral shock, ξ_{dt}^{nm} . The bilateral shock enters with opposite signs for the two countries.¹⁹ Specifically, for $n > m$,

$$\ln \xi_{mnt} = \ln \xi_{ct} + 0.5 \ln \xi_{dt}^{nm} \quad (4)$$

$$\ln \xi_{mnt} = \ln \xi_{ct} - 0.5 \ln \xi_{dt}^{nm} \quad (5)$$

where both the common and differential trade cost shocks follow an AR(1) process,

$$\begin{aligned} \xi_{ct} &= \bar{\xi}_c e^{\hat{\xi}_{ct}}, & \hat{\xi}_{ct} &= \rho_{\xi_c} \hat{\xi}_{ct-1} + \varepsilon_{\xi_{ct}}, & \varepsilon_{\xi_{ct}} &\sim N(0, \sigma_{\xi_c}), \\ \xi_{dt}^{nm} &= \bar{\xi}_d e^{\hat{\xi}_{dt}^{nm}}, & \hat{\xi}_{dt}^{nm} &= \rho_{\xi_d} \hat{\xi}_{dt}^{nm} + \varepsilon_{\xi_{dt}^{nm}}, & \varepsilon_{\xi_{dt}^{nm}} &\sim N(0, \sigma_{\xi_d}). \end{aligned}$$

Note that all the differential trade costs, $\xi_{mnt} \forall n > m$ share the same persistence, ρ_{ξ_d} and volatility, σ_{ξ_d} . There is no trade cost within a country, i.e. $\xi_{mnt} = 1$ for $n = m$.

There are two international financial frictions, $\{\phi_{nt}, F_t\}$. The interest rate shock, ϕ_{nt} , of

¹⁹It is straightforward to add a bilateral common shock process to account for bilateral trade agreements.

country n is country-specific and follows an AR(1) process

$$\phi_{nt} = e^{\hat{\phi}_{nt}}, \quad \hat{\phi}_{nt} = \rho_{\phi} \hat{\phi}_{nt-1} + \varepsilon_{\phi_{nt}}, \quad \varepsilon_{\phi_{nt}} \sim N(0, \sigma_{\phi}). \quad (6)$$

The debt elasticity, F_t , is common to all countries, and it follows an AR(1) process around the steady state $\bar{F}$,

$$F_t = \bar{F} e^{\hat{F}_t}, \quad \hat{F}_t = \rho_F \hat{F}_{t-1} + \varepsilon_{Ft}, \quad \varepsilon_{Ft} \sim N(0, \sigma_F).$$

A country n 's productivity shock is given as

$$\ln z_{nt} = \ln \bar{z}_n + \ln z_{ct} + \ln z_{dnt}.$$

It evolves around its steady state, $\bar{z}_n$, and is subject to the global shock, z_{ct} , and country-specific shock, z_{dnt} , all which follow an AR(1) process,

$$\begin{aligned} \ln z_{ct} &= \rho_{zc} \ln z_{ct-1} + \varepsilon_{zct}, \quad \varepsilon_{zct} \sim N(0, \sigma_{zc}) \\ \ln z_{dnt} &= \rho_{zd} \ln z_{dnt-1} + \varepsilon_{z_dnt}, \quad \varepsilon_{z_dnt} \sim N(0, \sigma_{zd}). \end{aligned}$$

5.2 Assigned and matched parameters

We divide the parameters into two groups. The first is set externally, and the second is estimated jointly to match the relevant cross-section and time-series moments. Table 2 reports the parameter values.

The first group includes the discount factor β , capital share α , depreciation rate δ_k , the intertemporal elasticity of substitution $1/\sigma$, the consumption weight μ , the preference weights on foreign goods ω_{mn} , the average debt $\bar{b}_n$, the lag in import demand δ , and persistence of the country-specific productivity shock ρ_{z_d} . Because the model is annual, we set $\beta = 0.96$ to generate an annual interest rate of 4%. The capital share α is 0.36, which is consistent with the labor share in the U.S. The depreciation rate δ_k is 10% annually. We set the intertemporal elasticity of substitution $1/\sigma$ as 0.5, which implies a standard risk aversion of

2. The weight on consumption μ is set to 0.37 to get the steady state hours of 1/3. We assume no home bias and set $\omega_{nm} = 0.25 \forall n, m$.²⁰ The steady state asset level $\bar{b}_n$ equals zero for all countries. The lag in import demand δ is set to 0.52 to generate gradualness in trade flows.²¹ This is the highest level of δ that preserves determinacy in the model.²² When δ equals zero, we have the standard model with no trade adjustment dynamics. We set the persistence of the common productivity shocks as $\rho_{zc} = 0.96$ and conduct sensitivity analysis on them (Appendix D.1).

The 17 parameters in the second group are chosen to match the model's moments to the data. This includes the Armington elasticity γ , the RER elasticity in pricing-to-market θ , the debt elasticity of interest rate at the steady state $\bar{F}$, volatility and persistence of financial shocks $\sigma_F, \rho_F, \sigma_\phi, \rho_\phi$, volatility and persistence of trade shocks $\sigma_{\xi_c}, \rho_{\xi_c}, \sigma_{\xi_d}, \rho_{\xi_d}$, as well as the steady state values of common trade cost and productivity $\bar{\xi}_c, \bar{z}_n$. We first set the mean values $\bar{\xi}_c$ and $\bar{z}_n$ to generate the average observed trade share over output and the relative GDP of the countries, respectively. Then we choose the remaining parameters jointly with patterns of cross-country dispersion with trade, as well as the country-level business cycle moments. The seven cross-country moments that relate to trade are the regression coefficients of TBY, export-import ratio, real exchange rate, and terms of trade in relation to TRY.²³ We also target the long-run elasticity of trade. For the business cycle moments, we focus on a set of three moments regarding the volatility of output, consumption, and investment. We consider standard open economy moments including the Backus-Smith condition and business cycle comovement. We also target the moments related to TRY, including the autocorrelation and volatility of TRY. The moments we target to pin down the parameters are presented in Table 3 along with the untargted moments.

We solve our model using perturbation methods. As we are interested in the effect of shocks that alter the structure of business cycles, we require a second-order approximation

²⁰Alternatively, we can introduce home bias and adjust the mean trade cost accordingly. The results remain essentially unchanged.

²¹The input adjustment cost δ determines the differences between short- and long-run trade elasticities.

²²Corsetti et al. (2008) show that international macro models may have multiple equilibria with incomplete financial markets, persistent shocks, and a low Armington elasticity.

²³We estimate the regression of TBY through the origin as the absence of trade implies zero dispersion in TBY.

of the model. With many countries, this greatly increases computational burden.²⁴ We construct the moments based on a 10,000 year simulation of the model. The cross-sectional moments are based on 50 year windows of our simulations. We take an indirect inference approach to identify the parameters, by choosing them to minimize the distance between the model moments and the data.

Every parameter matters for the general equilibrium and affects all the moments. However, there is a clear correspondence between certain parameters and moments. The Armington elasticity γ disciplines the response of prices and matters for the comovement of integration and global dispersion in the trade balance. The resulting γ is 3.22. The pricing-to-market parameter θ governs the relative volatility of the terms of trade and real exchange rate. When θ equal zero, we have the standard model with complete pass-through. The estimation calls for positive values: $\theta = 0.53$. Both the financial friction, governed by debt elasticity $\bar{F}$ and the UIP wedge ϕ , and trade barrier, ξ_c , affect cross-border capital flows. A higher debt-elasticity $\bar{F}$ reduces intertemporal risk-sharing and lowers the volatility of the trade balance. It also allows the model to match the observed dispersion in export-import ratio. Its estimated steady state level is $\bar{F} = 2.01$, which is substantially larger than the value used in standard models with constant debt elasticity. Its volatility is $\sigma_F = 0.09$, and persistence is $\rho_F = 0.99$. The process for the common trade barrier ξ_c governs the volatility and the autocorrelation of gross trade. The estimated common trade shock is $\sigma_{\xi_c} = 0.02$ and $\rho_{\xi_c} = 0.997$. While global trade shocks are slightly less volatile than global changes in financial frictions, the persistent behavior of gross trade flows calls for high persistence of global trade shocks. The process for the differential trade barrier ξ_d matters for the volatility of the real exchange rate as well as consumption. We find that $\sigma_{\xi_d} = 0.03$ and $\rho_{\xi_d} = 0.65$. Common and differential productivity are related to the volatility of GDP and its cross-country correlation, respectively. Each of their volatility is $\sigma_{z_c} = 0.02$ and $\sigma_{z_d} = 0.02$. The capital adjustment cost parameter, ψ , is pinned down by the volatility of investment, which result in the value of $\psi = 121.82$.

Note that in our benchmark estimation, we let the debt elasticity F , UIP wedge ϕ , and

²⁴The number of state variables increases rapidly with the number of countries. Specifically, it follows $\frac{3}{2}n(n-1)+6n-1$ with n countries, implying quadratic growth in n . We focus on 8 countries in the benchmark analysis, which gives 143 state variables, and later conduct robustness to the number of countries.

trade cost ξ_c vary over time, following the estimated process. For example, a global shock to the debt elasticity parameter captures periods of high and low costs to international borrowing and lending. We then consider counterfactual cases when there are no changes in financial or trade frictions. We discuss the results and the features of the model delivering these results in the following sections.

5.3 Model Results

The parameters in our model are chosen to match key features of the distribution of economic activity in the global economy over time. Specifically, we target data moments related to trade integration, trade balance dynamics, international relative prices, and business cycles. Table 3 reports both the key data moments and the corresponding model-implied moments. The top panel summarizes the relationship between cross-country dispersion and integration moments. Figure 7 visualizes the relationship with a scatter plot, with colored circles representing data and black crosses showing the model predictions.²⁵

The first panel of Figure 7 shows the model reproduces the strong positive relationship between the standard deviation of TBY and trade openness as in data. The positive slope indicates that the net trade flows are more dispersed as the gross trade grows. By contrast, the upper-right panel shows that dispersion in the export-import ratio decreases with higher gross trade. This occurs because lower trade costs promote risk sharing and the reallocation of capital across countries. This leads to more dispersed net trade flows and aligned movement of exports and imports. Furthermore, the remaining four panels in Figure 7 show that the model successfully captures the decomposed components of the export–import ratio. The middle panels indicate that higher economic integration reduces the dispersion of relative prices across countries, consistent with greater price convergence under trade openness. The lower panels show that relative spending becomes less dispersed as the world economy becomes more integrated, in both the data and the model. For relative output, the slopes remain close to zero in both.

²⁵In the quantitative analysis, we study TBY in nominal terms and the export–import ratio in real terms to better discipline price movements. However, conducting the analysis entirely in nominal or real terms would make little difference to the results, as our exercise focuses on matching relative movements in prices and quantities.

We also consider several major business cycle moments, reported in the second panel of Table 3. Overall the model is a good fit. It matches the volatility of output, consumption, and investment by targeting them. The volatility and persistence of TRY are slightly lower in the model than in the data, suggesting that trade shocks would need to be even more persistent than in our current parameterization.²⁶ The model also reproduces the observed international comovement in output and the weak correlation between TBY and output, consistent with international borrowing and lending cycles. Furthermore, it successfully account for the departures from the Backus–Smith condition, with the negative correlation between relative consumption and the real exchange rate. Finally, the model produces the persistence of output and consumption that are very close to those observed in the data, although they are not directly targeted. The last panel of Table 3 reports the trade elasticity, showing a higher value in the long run than in the short, consistent with gradual adjustment of trade.

Finally, Figure 8 plots the model-generated moments across countries, where each point represents one of the eight countries in the model. These correspond to the empirical cross-country moments shown in the lower panels of Figures 2 and 4, which are also plotted here in blue circles for reference. Note that these model moments are not targeted in the calibration. Although the model includes fewer countries, it replicates the data patterns remarkably well. The positive and negative elasticities of TBY and the export–import ratio with respect to trade closely match those observed in the data.

Financial Integration and Global Dispersion We now explore the role of changes in international financial frictions on the aggregate behavior of our model economy. To do so, we hold the financial frictions constant over time by eliminating shocks to the debt elasticity parameter ($\sigma_F = 0$). Results are reported in the column "No financial integration" of Table 3 and in Figure 9.

Removing financial integration shocks lowers the elasticity of the TBY dispersion and trade slightly, from 0.09 to 0.08. It does lower dispersion in real net trade flows, measured by

²⁶Under the AR(1) process for the common trade shock, our estimated persistence is 0.997. We later consider adding a trend shock to trade costs and show that the results are robust to this alternative process in Section 5.4.

the constant in our export-import regression, by about 10 percent and brings the elasticity of dispersion in the export-import closer to zero and thus further from the data. These changes in the nature of borrowing with trade largely reflect a smaller decline in relative price dispersion with trade.

Trade Integration and Global Dispersion We now discuss how global trade reforms influence the distribution of business cycles across countries. We eliminate shocks to the global trade shocks ($\sigma_{\xi_c} = 0$). Results are reported in the column "No trade integration" of Table 3 and in Figure 9. Without these common trade shocks, there are very minor changes in gross trade flows, resulting in a large decrease in the volatility and persistence of TRY from 0.08 to 0.01 and from 0.70 to 0.37, respectively. More importantly, we lose the significant relationship between net trade flows and gross trade. As shown in the figure, the relationship between trade and net trade dispersion is almost vertical, making the slope of the cross-country regression excessively high. Also, in the absence of common trade shocks but with financial shocks, the dispersion in both of the export-import ratio becomes excessively large. This is consistent with [Mac Mullen and Woo \(2025\)](#) that highlighted financial shocks alone generate excessive movements in net trade flows. Hence, the model without trade integration accounts for very little of how our measures of dispersion vary with world trade.

5.4 Robustness

We next examine how key modeling assumptions affect our results. We first show that the results are robust to expanding the asset structure. We then evaluate the roles of pricing-to-market, trade adjustment costs, the stochastic process for trade costs, the number of countries, and the order of approximation used to solve the model. In each exercise, we alter specific parameters without recalibrating the rest of the model. Across these variations, our main findings remain robust.

Financial Market We show that allowing for a richer set of assets yields a similar qualitative relationship between trade integration and the changing nature of net trade flows. Specifically, we introduce a complete set of contingent claims into our model. With this asset

structure, we remove the UIP shocks. Now the model includes the familiar Backus-Smith condition, $u_{ci}/P_i = \lambda_i u_{c1}/P_1$, where λ_i is set so that in steady state each country has balanced trade. The remaining parameters are the same as in our baseline calibration. The properties of the model are reported in the columns "Complete market" in Table 3 and Figure D.1.

While the model with complete markets is a worse fit to the data, it yields similar qualitative relations between net flows and gross flows over time. Introducing complete markets increases net trade flows overall and the elasticity of the dispersion in TBY and export-import ratio with TRY. In terms of business cycles and international transmission, we find that output, consumption and investment are a bit smoother, and as we have mentioned the Backus-Smith condition now holds with the correlation equal to one.

As before, shutting off the trade shocks yields a large positive relationship between dispersion in the export-import ratio and overall trade (Figure D.1).²⁷ It also leads to a counterfactual increase in the dispersion of relative expenditures and relative output with trade. The findings here suggest that changes in financial frictions that bring the world closer to the complete markets allocation should operate much like our model with a non-contingent bond with a falling debt elasticity parameter.

No PTM To capture the movements in terms of trade and real exchange rate, we introduce a pricing-to-market (PTM) friction. We report the results of eliminating this friction ($\theta = 0$) in the column "No PTM" of Table 3 and Figure D.2. The model is a worse fit to the data on several moments, but our main focus on the relationship between net trade and trade integration is unchanged. However, the PTM friction is needed for the terms of trade to be less volatile than the real exchange rate as in the data. Absent this friction, the terms of trade become as volatile as the real exchange rate, making the terms of trade much more volatile than in the data. Correspondingly, the model plots lie well above the data points. Note that in the benchmark case where the friction is introduced, the model and data plots closely overlap (Figure 7).

²⁷For regression moments that correspond to the graph with and without trade integration, see Table D.3.

No trade adjustment frictions Trade takes time to adjust to shocks that move the exchange rate or trade barriers. We capture this feature by introducing a form of external habit in the aggregator. We show that these trade adjustment frictions dampen net trade flows over the business cycle. We now shut this force off by setting $\delta = 0$. Results are shown in the column "No trade adjustment" of Table 3 and Figure D.3. Without the adjustment friction the short-run trade elasticity now is similar to the long-run trade elasticity. Aside from leading trade to adjust faster, there are modest changes in the model's business cycle properties. Business cycles are a bit more volatile with output, consumption, and investment volatility growing. Importantly, the elasticity of dispersion in TBY grows more strongly with the level of world trade than our baseline from 0.09 to 0.14. Moreover, the average level of dispersion in the export-import ratio increases substantially from 0.52 to 0.58, so it is easier to borrow and lend.

Gradual trade reforms We focus on trade integration through a mean-reverting process of global trade costs. In practice, most multilateral or regional trade agreements involve phasing out tariff and non-tariff barriers in several steps.²⁸ To capture these reforms, we redo our analysis with an alternative process for the common trade costs that captures these phaseouts. These phaseouts lead to stronger role for trade integration in changing the nature of borrowing and lending than our baseline.

To generate persistent movements in trade growth from trade policy, we add a trend shock to the trade cost, ξ_c :

$$\begin{aligned}\xi_{ct} &= (1 - \rho_{\xi_c}) \cdot \bar{\xi}_c + \rho_{\xi_c} \cdot \xi_{ct-1} + \Delta_t + \varepsilon_{\xi_{ct}} & \varepsilon_{\xi_{ct}} &\sim N(0, \sigma_{\xi_c}), \\ \Delta_t &= \rho_{\Delta} \cdot \Delta_{t-1} + \varepsilon_{\Delta t} & \varepsilon_{\Delta t} &\sim N(0, \sigma_{\Delta}).\end{aligned}$$

For this case, we set $\rho_{\Delta} = 0.75$. We eliminate the transitory common shocks, $\sigma_{\xi_c} = 0$, and set the trend shock to match the volatility of TRY, $\sigma_{\Delta} = 0.01$. This captures the persistent movements of trade costs following phased-in reforms. We simulate the model as in the

²⁸For instance, the 1967 GATT round had 5 annual steps, the 1980 GATT round had 7 annual steps, and the 1995 GATT round had 6 annual steps. Regional trade agreements like the European Community, CUSFTA, and NAFTA were similar.

benchmark case, but this time we focus on the realizations of shocks that induce world trade to increase as in the data. That is, we compute moments from specific 50-year periods in which the TRY increase from under 30 percent at the beginning of the period to above 50 percent by the end.

The results are shown in the column "Phased-in reforms" of Table 3. The model is a slightly better fit in trade dynamics. We now match the exact volatility of TRY, with $\sigma(TRY) = 0.15$ and generate a persistence of TRY that is much closer to the data. The relationships between trade integration and the dispersion in net trade and relative prices remain robust, although the elasticities are slightly weaker than in the benchmark case. These relationships also continue to hold in the time-series moments. Other moments of business cycle are mostly unchanged with this process of trade reforms.

Number of countries We use eight countries in our baseline because the second-order solution becomes increasingly costly as the number of countries grows. To test robustness, we simulate a 10-country version of the model using the same parameters. The results closely mirror the baseline: dispersion remains tightly linked to trade openness, and removing trade integration again produces excessive volatility in net trade flows. For more details, see Appendix D.2.

The result are depicted in Figure D.4 and in Table 3 Column "10 countries." This model yields similar moments to our baseline 8 country model. In particular, the dispersions are slightly less elastic to trade. In any case, however, most of the moments are very close to the baseline case.

We also run counterfactual exercises without financial or trade integration. The model results are again consistent with our baseline case. Without trade integration, the net trade gets excessively volatile and there is not enough increase in the gross trade (red triangles in Figure D.4). Turning off financial integration has a minor effect.

Computational Approach We now show that alternative computation strategies yield qualitatively similar results but miss out on some interactions between integration and business cycles. Recall that the estimated relationships between the dispersion of net flows and

trade integration in the data are from a transition of a less integrated world to a more integrated world. To capture this transition requires solving the model with either a global solution or a high-order approximation. Alternatively we can solve the model with a first-order approximation around different steady states with different levels of trade integration. That is, we combine the results from two simulations, each with high and low level of steady state common trade cost, $\bar{\xi}_c$. We also use half the size of the common trade shock σ_{ξ_c} within each simulation. Aside from these trade parameters, the remaining parameters are the same as in our baseline case.

Figure 10 and column "First order" show the results. We find that the qualitative relationships of the benchmark model continue to hold, but the slopes are much weaker. For instance, the positive relationship between trade openness and the TBY dispersion remains, but the magnitude of the slope is reduced. The export-import ratio has a negative slope, although it is flatter than in the baseline model or the data. Similarly, the real exchange rate and the terms of trade patterns are preserved but dampened. Moreover, the graph shows that the overall cross-country dispersion becomes smaller under the first-order approximation. The dispersion in TBY, export-import ratio, the real exchange rate, and the terms of trade lie lower on the graphs compared with the benchmark model. The last two panels, which report relative spending and output, show relatively little sensitivity to the order of simulations.

When we focus on the behavior within simulations around each steady state, the sensitivity of dispersion to trade openness becomes even weaker. In fact, within a given steady state, the dispersion in the export-import ratio is essentially flat across different levels of trade when the model is solved using a first-order approximation. The same pattern appears for the real exchange rate and the terms of trade. Only when observations from the low-trade and high-trade steady states are pooled together do we recover a negative slope. We attribute the stronger relationship to two features. First, common trade shocks will have asymmetric effects across countries owing to openness²⁹ and second, trade integration changes macroeconomic volatility through its interaction with other shocks.

²⁹These asymmetric effects of global shocks are emphasized in [Alessandria et al. \(2017\)](#), [Barattieri \(2014\)](#), [Ravikumar et al. \(2019\)](#) and [Ju et al. \(2021\)](#).

6 Summary

We contribute to a growing literature that interprets international time series on business cycles and trade integration through a dynamic stochastic general equilibrium model. This literature departs from conventional approaches that treat trade and unbalanced trade as separate phenomena studied in isolation. Our approach allows us to study the changing nature of business cycles, economic integration, and the spatial allocation of economic activity. We argue that the rise in the international trade and cross-border borrowing and lending from 1970 to 2019 has a common cause: the decline in policy and technological barriers to moving goods across countries. The changing nature of business cycles with trade is also consistent with this view.

We develop a multi-country variation of the canonical multi-good RBC model of [Backus et al. \(1994\)](#) with the usual business-cycle shocks plus shocks to international trade barriers and financial frictions. Relating the model to the data, we show that most of the rise in cross-country borrowing and lending over time is driven by declining international trade barriers. With lower trade barriers, it becomes easier to borrow and lend in response to shocks without inducing a larger movement in the real exchange rate or terms of trade, consistent with the data. We find little evidence that cross-border financial frictions have decreased or that countries are experiencing more asymmetric shocks. Indeed, these alternative explanations would imply an increase in the dispersion of relative prices and net trade flows as a share of overall trade, rather than the moderation in the data.

We interpret the data through a model in which financial frictions between countries have no direct effect on the level of trade, and the level of trade has no direct effect on cross-border financial frictions. There is mixed evidence on the relationship between financial development and trade ([Beck, 2003](#), [Leibovici, 2021](#)), but we view this evidence to be too weak for this margin to explain much of the growth in trade. More likely, trade could affect financial frictions, as borrowing capacity could be tied to debt relative to exports or the level of trade rather than output. More work should relate financial and trade integration in dynamic stochastic general equilibrium models and discipline the models predictions with data over long periods with structural changes in these international relations.

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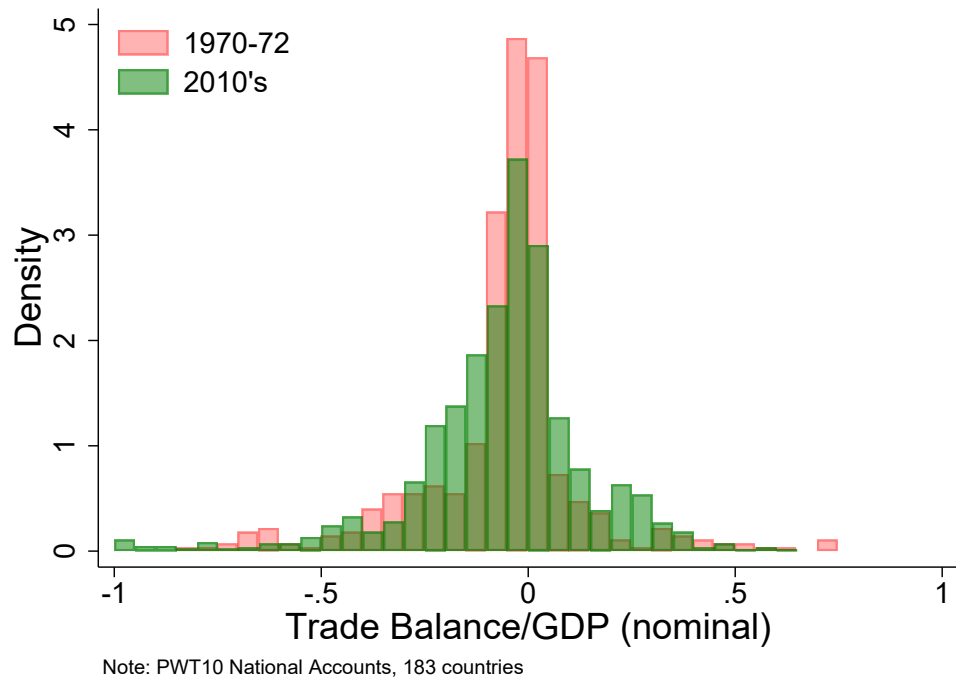
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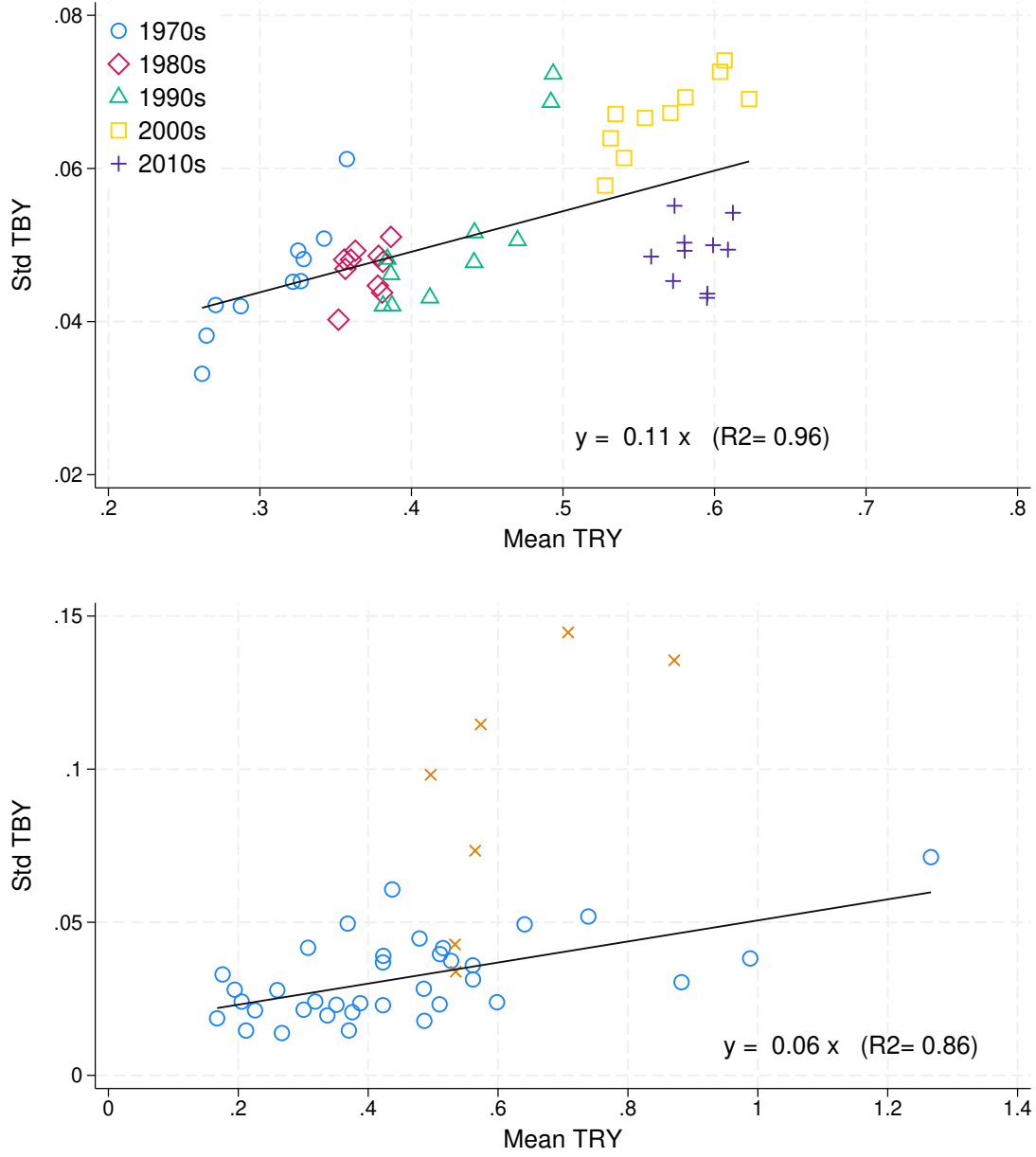
FIGURES AND TABLES

Figure 1: Distribution of Trade Balance: 1970s vs 2010s



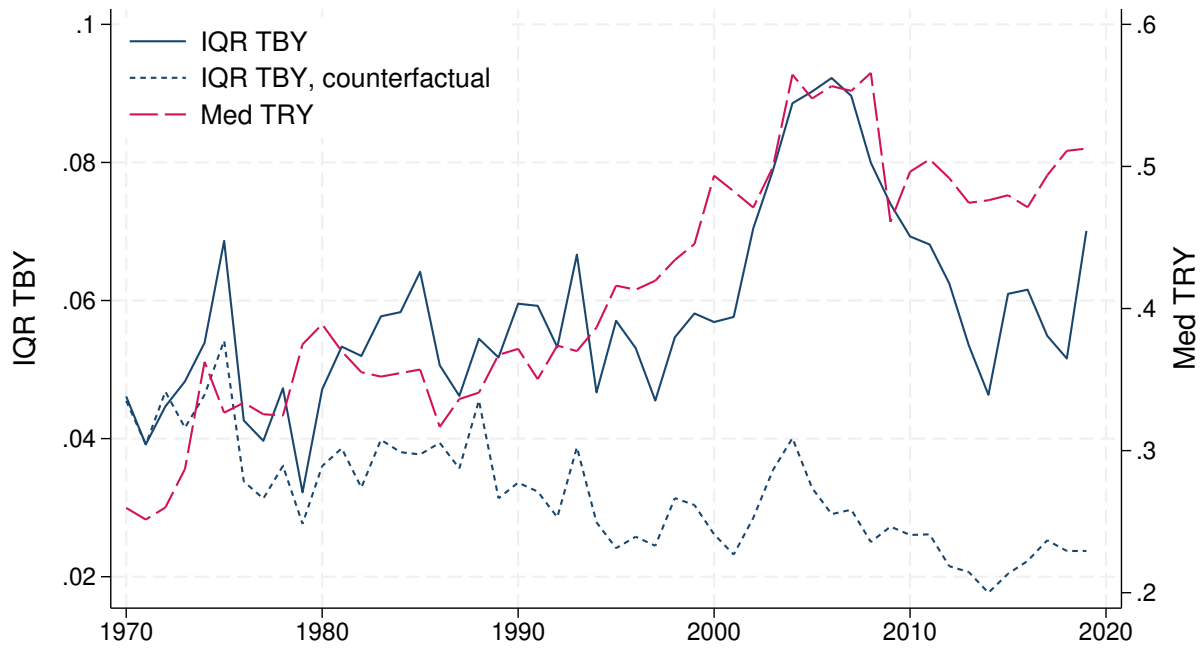
Notes: Comparison of the distribution of the trade balance as a share of GDP between the 1970-1972 and 2010-2019, measured in nominal terms. The trade balance as a share of GDP equals $TBY_n = (X_n - M_n)/Y_n$ and is based on 183 countries from the Penn World Tables version 10 National Accounts data.

Figure 2: Trade Balance Dispersion and Trade



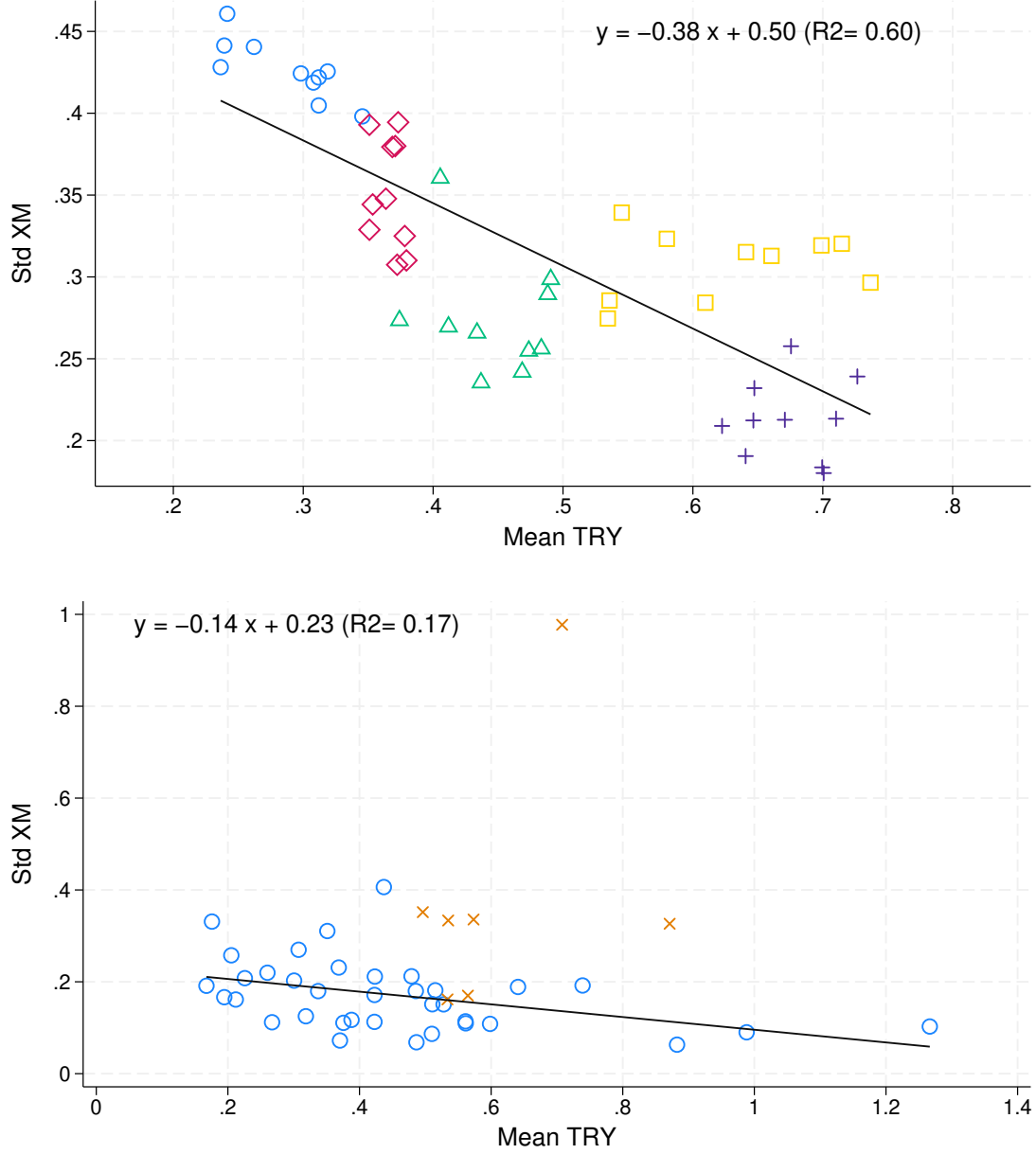
Notes: The upper panel plots cross-country moments for each year, with decades presented in different colors. The lower panel shows the time series moments for each country. The x-markers in the lower panel represent countries with measurement concerns that are not included in our baseline sample and are not taken into account for the regression. The variables are in nominal terms.

Figure 3: Trade Balance Dispersion and Trade over Time



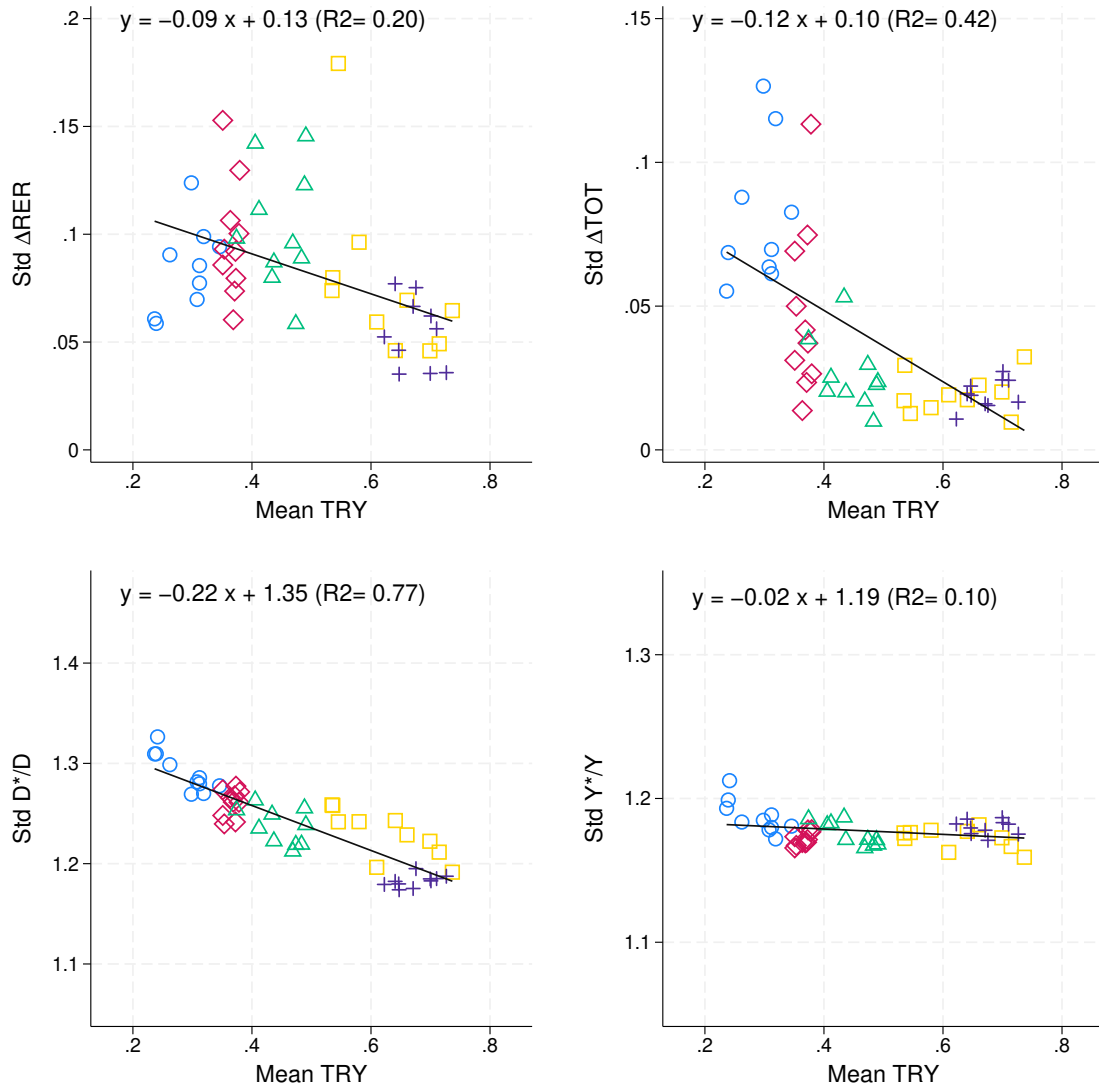
Notes: The figure plots interquartile range of TBY median of TRY. The counterfactual TBY is calculated by fixing the TRY at the 1971 level for each country. The moments are calculated based on our baseline sample of 36 countries.

Figure 4: Export-Import Ratio Dispersion and Trade



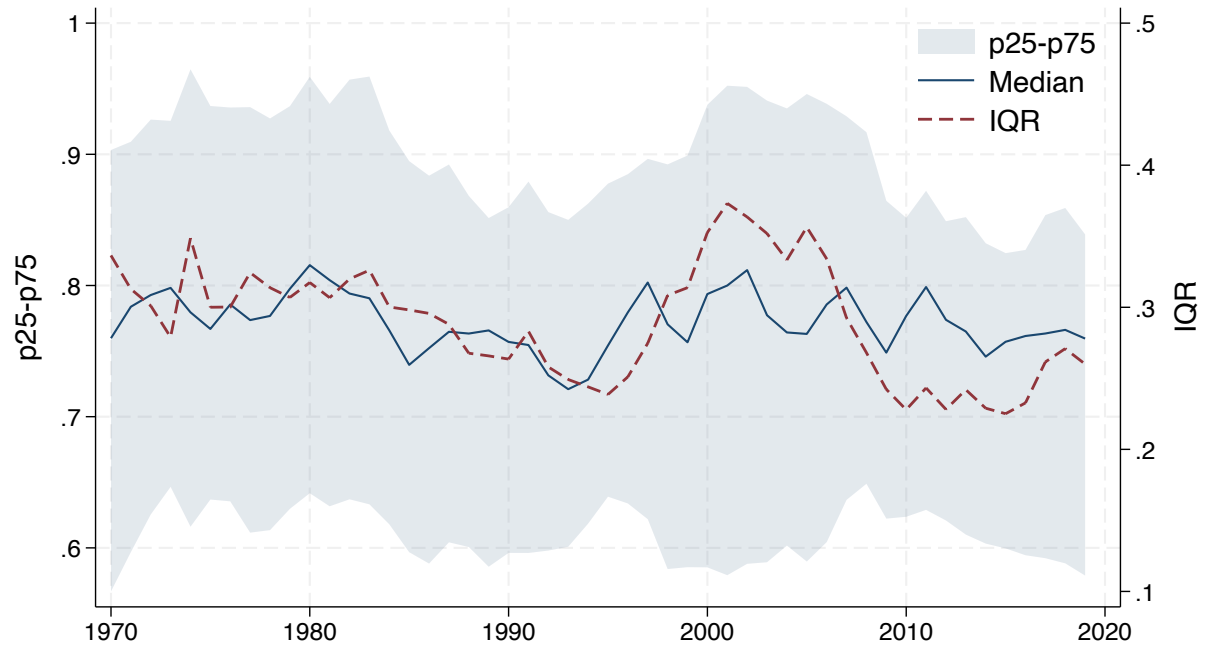
Notes: The upper panel plots cross-country moments for each year, with decades presented in different colors. The lower panel shows the time series moments for each country. The x-markers in the lower panel represent countries with measurement concerns that are not included in our baseline sample and are not taken into account for the regression.

Figure 5: Macroeconomic Dispersion and Trade



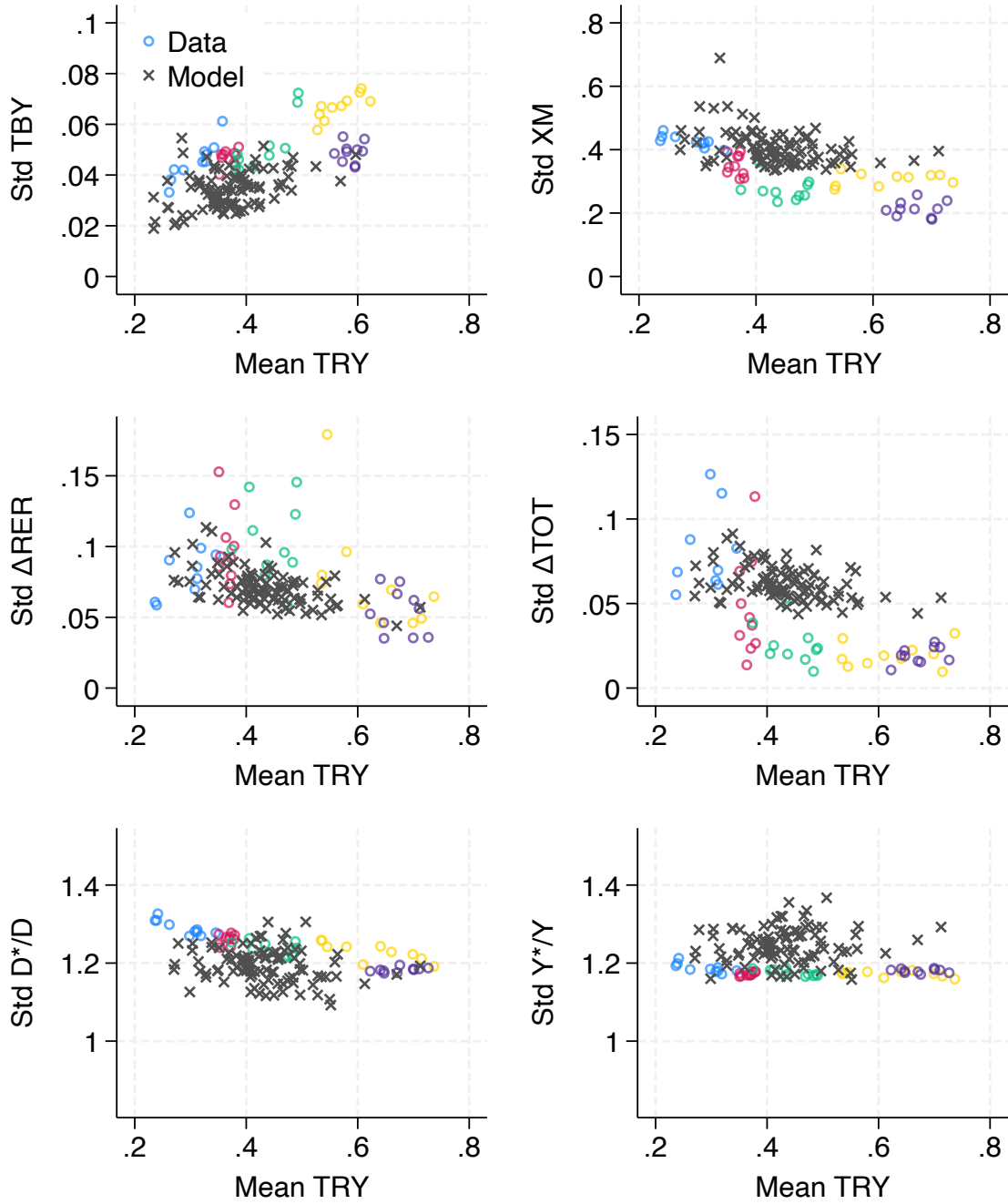
Notes: The graphs plot cross-country moments for each year, with decades presented in different colors. The dispersion is measured by the standard deviation of each variable.

Figure 6: TFP Dispersion



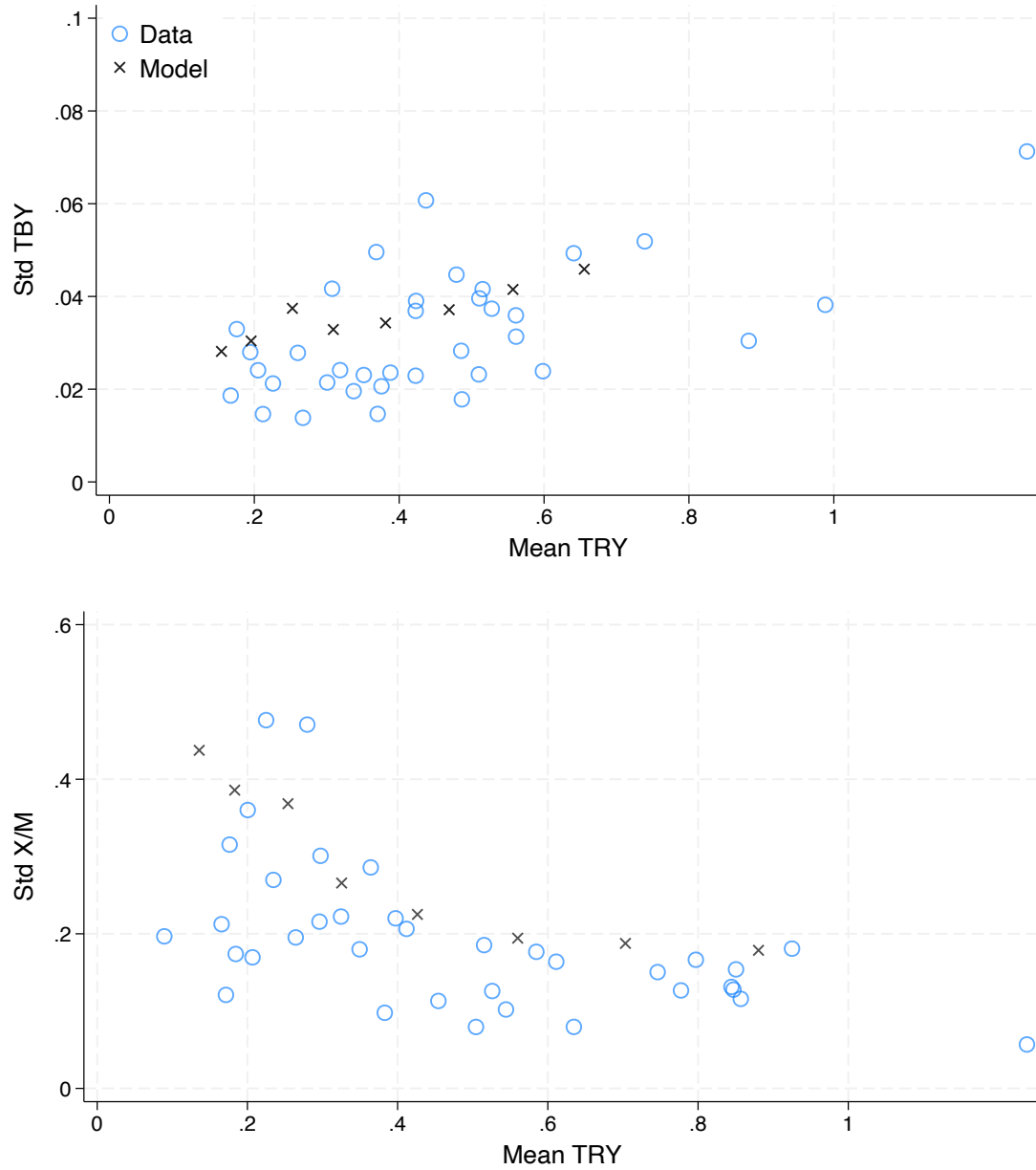
Notes: The graph plots the dispersion in TFP at current PPPs across countries in our baseline sample. The shaded area on the left axis represents the range across the sample, with the upper and lower bounds corresponding to the 75th and 25th percentiles, respectively. The red dashed line on the right axis shows the size of this range, or the interquartile range (IQR), over time.

Figure 7: Model Results on Global Dispersion and Trade



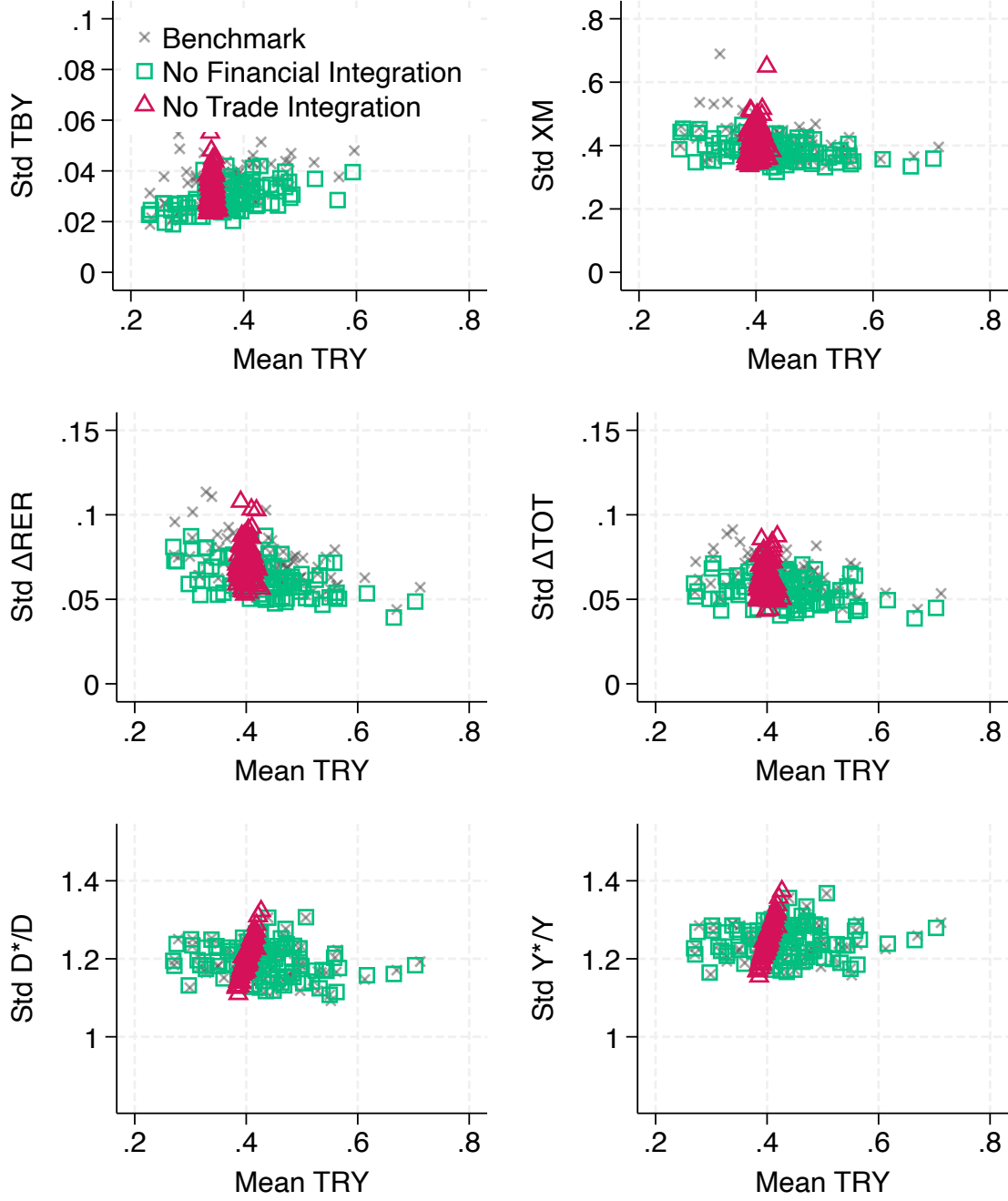
Notes: In each graph, y-axis is the standard deviation, x-axis is trade share over output. In the first panel, TBY and TRY are measured in nominal terms, while the rest of the panels are in real terms. The colored dots show the data, and the black crosses show the baseline model prediction.

Figure 8: Model Results across Countries



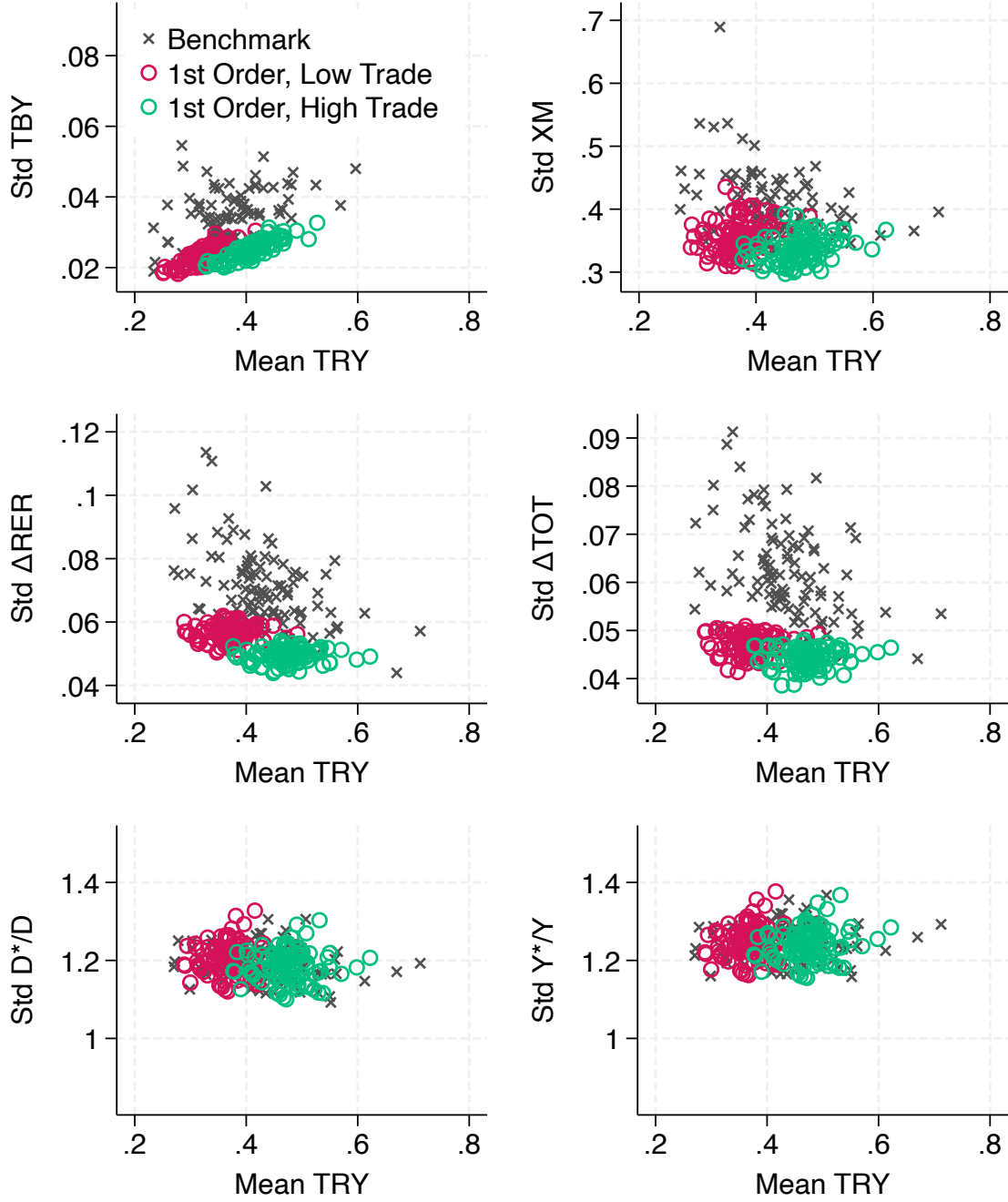
Notes: In each graph, y-axis is the standard deviation, x-axis is trade share over output. The colored dots show the data, and the black crosses show the baseline model prediction.

Figure 9: No Integration



Notes: In each graph, y-axis is the standard deviation, x-axis is trade share over output. The colored dots show the data, and the black crosses are when there is no financial integration, or $\sigma_F = \sigma_\phi = 0$.

Figure 10: Sensitivity to Computational Approach



Notes: In each graph, y-axis is the standard deviation, and x-axis is trade share over output. The model results are based on a first order approximation around two different levels of openness. The black line shows the model results.

Table 1: Estimation of Armington Elasticity

	Level		Difference		ECM	
	(1)	(2)	(3)	(4)	(5)	(6)
SR price	0.0588 (0.0587)	0.107*** (0.0203)	0.234*** (0.0355)	0.232*** (0.0360)	0.244*** (0.0357)	0.237*** (0.0359)
SR spending	$I^\dagger$	0.00121 (0.00578)	$I^\dagger$	1.019*** (0.0720)	$I^\dagger$	1.058*** (0.0743)
Adjustment					0.00614** (0.00218)	0.00621** (0.00218)
LR price					3.306* (1.551)	3.455* (1.595)
Observations	1750	1750	1715	1715	1715	1715
R-squared	-10.66	0.0255	0.277	0.278	0.286	0.286

Notes: Data from Penn World Table 10.0, 1970-2019 with 36 countries. Superscript $\dagger$ denotes that the coefficient is not estimated but is set to 1 as the theory suggests. ECM stands for the error correction model: $\Delta \ln X_t/M_t = \beta + \gamma_{SR}\Delta p_t^r + \Delta d_t^r - \alpha(\ln X_{t-1}/M_{t-1} - \gamma_{LR}p_{t-1}^r - d_{t-1}^r)$. Robust standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 2: Calibrated Parameters

Parameters		Value
Externally Calibrated		
β	Discount factor	0.96
α	Capital share	0.36
δ_k	Capital depreciation rate	0.10
$1/\sigma$	Intertemporal elasticity of substitution	0.50
μ	Weight on consumption	0.37
ω_{mn}	Weight on each foreign good	0.25
ω_{nn}	Weight on home goods	0.25
$\bar{b}_n$	Mean debt	0
δ	Trade habit	0.52
ρ_{z_c}	Common productivity persistence	0.96
$\bar{\xi}_d$	Mean differential trade cost	1
Internally Calibrated		
$\bar{\xi}_c$	Mean common trade cost	5.45
$\bar{z}_n$	Mean productivity	$0.72^n, n \in \{0, 1, 2, \dots, 7\}$
γ	Armington elasticity	3.22
θ	RER-elasticity of PTM	0.53
ψ	Capital adjustment cost	121.82
F	Debt-elasticity of interest rate	2.01
σ_F	Debt-elasticity shock volatility	0.09
ρ_F	Debt-elasticity shock persistence	0.99
σ_ϕ	Interest rate shock volatility	0.05
ρ_ϕ	Interest rate shock persistence	0.99
ρ_{ξ_c}	Common trade cost persistence	0.997
σ_{ξ_c}	Common trade cost shock volatility	0.02
ρ_{ξ_d}	Differential trade cost persistence	0.65
σ_{ξ_d}	Differential trade cost shock volatility	0.03
σ_{z_d}	Differential productivity volatility	0.02
σ_{z_c}	Common productivity volatility	0.02
ρ_{z_d}	Differential productivity persistence	0.98

Table 3: Model Results

		Data	Baseline	No financial integration	No trade integration	Complete market	No PTM	No trade adjustment	Phased-in reforms	10 countries	First order
TBY	slope	0.11	0.09 [†]	0.08	0.09	0.24	0.10	0.14	0.08	0.08	0.07
X/M	slope	-0.38	-0.25 [†]	-0.17	1.14	-0.59	-0.33	-0.31	-0.06	-0.12	-0.07
X/M	constant	0.50	0.52 [†]	0.46	-0.05	0.85	0.57	0.58	0.41	0.41	0.38
ΔRER	slope	-0.09	-0.08 [†]	-0.06	0.04	-0.02	-0.08	-0.06	-0.03	-0.06	-0.04
ΔRER	constant	0.13	0.11 [†]	0.09	0.06	0.02	0.10	0.09	0.08	0.10	0.07
ΔTOT	slope	-0.12	-0.05 [†]	-0.03	-0.03	0.00	-0.09	-0.04	-0.01	-0.03	-0.02
ΔTOT	constant	0.10	0.08 [†]	0.07	0.08	0.01	0.14	0.07	0.06	0.07	0.05
D^*/D	slope	-0.22	-0.12 [†]	-0.12	4.22	-0.04	-0.11	-0.17	-0.17	-0.26	-0.11
D^*/D	constant	1.33	1.24 [†]	1.25	-0.49	1.22	1.24	1.27	1.27	1.17	1.24
Y^*/Y	slope	-0.01	0.03 [†]	0.02	4.51	0.11	0.03	-0.03	-0.02	-0.14	0.03
Y^*/Y	constant	1.17	1.23 [†]	1.24	-0.56	1.21	1.23	1.26	1.26	1.16	1.23
Integration and Time-series Volatility											
TBY	slope	0.06	0.09	0.07	0.09	0.22	0.09	0.14	0.08	0.08	0.05
X/M	slope	-0.14	-0.35	-0.32	-0.38	-0.23	-0.36	-0.19	-0.24	-0.20	-0.20
X/M	constant	0.23	0.43	0.38	0.44	0.66	0.47	0.43	0.33	0.34	0.27
Business Cycle Moments											
$\sigma(\Delta y)$, %		3.69	3.58 [†]	3.41	3.56	3.23	3.74	3.81	3.50	3.66	3.23
$\sigma(\Delta c)/\sigma(\Delta y)$		1.06	0.78 [†]	0.81	0.78	0.74	0.77	1.01	0.80	0.80	0.84
$\sigma(\Delta inv)/\sigma(\Delta y)$		3.16	3.18 [†]	2.92	3.08	1.70	3.62	5.01	3.45	3.40	2.40
$\rho(\Delta y_t, \Delta y_{t-1})$		0.39	0.21	0.25	0.22	0.33	0.17	0.19	0.22	0.20	0.31
$\rho(\Delta c_t, \Delta c_{t-1})$		0.37	0.37	0.38	0.37	0.58	0.35	0.13	0.35	0.36	0.40
$\sigma(TRY)$		0.15	0.08 [†]	0.08	0.01	0.06	0.08	0.08	0.15	0.08	0.04
$\rho(TRY)$		0.98	0.70 [†]	0.71	0.37	0.58	0.70	0.70	0.99	0.69	0.71
$\rho(TBY, \Delta y)$		-0.01	0.00	0.01	-0.01	-0.04	-0.02	0.39	0.00	0.00	0.00
$\rho(\Delta(c - c^*), \Delta RER)$		-0.21	-0.22 [†]	-0.20	-0.22	1.00	-0.28	-0.72	-0.13	-0.24	-0.07
$\rho(\Delta y, \Delta y^*)$		0.21	0.21 [†]	0.25	0.22	0.33	0.17	0.19	0.22	0.20	0.28
Trade Elasticity											
Short Run		0.24	1.54	1.54	1.54	1.57	1.54	3.17	1.55	1.54	1.53
Long Run		3.31	3.29 [†]	3.30	3.29	3.24	3.27	3.18	3.38	3.31	3.35

Notes: The superscript [†] denotes that the moment is targeted using the internally calibrated parameters. All of the moments except the slope for TBY are calculated using real values. For estimating the elasticity of TBY, both TBY and TRY are in nominal values and the constant is fixed to be zero. RER is the real exchange rate relative to the US, while TOT is the terms of trade calculated as the logged ratio of price level of exports to imports of each country, both in data and the model. Business cycle moments are averages across countries.

APPENDIX (NOT FOR PUBLICATION)

In this section we discuss five topics. First, Appendix A describes the data in our empirical analysis. Second, Appendix B shows that the positive relationship between the level of gross trade flows and the dispersion of net trade flows is robust across many measures of net flows and country coverage. Third, Appendix C undertakes a variance decomposition. Fourth, Appendix D.1 examines the sensitivity of our model to the persistence of the shocks to productivity. Fifth, in Appendix D.2 we show that the theoretical relationship from the model is robust to considering more countries.

A Data

In this section we describe our data sources, variables, and variable construction. We focus on countries that are available in the Penn World Table (PWT) 10.0 since at least 1970, are covered in the broad basket of BIS Effective Exchange Rates, and have data on bilateral trade since 1970 in the Direction of Trade Statistics (DOTS). Among the 49 countries that satisfy these criteria, we exclude six entrepot countries,³⁰ resulting in the sample of 43 countries. Then we further exclude seven countries with measurement concerns in TBY.³¹ This leaves us with 36 countries,³² which we use as our baseline sample of countries for the period of 1970-2019.

Using PWT 10.0, we construct variables including trade to output ratio (TRY) and trade balance to output (TBY) as follows:

- TRY, real: Sum of export and import shares at current PPPs
- TRY, nominal: Sum of export and import shares at current PPPs, inflated by the price level of export relative to the price level of output

³⁰Belgium, Luxembourg, Hong Kong, Singapore, and Malta

³¹Algeria, Bulgaria, Cyprus, Iceland, Ireland, Norway, Saudi Arabia, and the United Arab Emirates. For details, see Appendix B.

³²Argentina, Australia, Austria, Brazil, Canada, Chile, China, Colombia, Denmark, Finland, France, Germany, Greece, Hungary, India, Indonesia, Israel, Italy, Japan, Malaysia, Mexico, Netherlands, New Zealand, Peru, Philippines, Poland, Portugal, Republic of Korea, Romania, Spain, Sweden, Switzerland, Thailand, Turkey, United Kingdom, and United States.

- *TBY*, nominal: Difference of export and import shares at current PPPs, inflated by the price levels
- X/M : Logged ratio of export and import shares at current PPPs
- *RER*: Logged price level of output relative to the US
- *TOT*: Logged ratio of the price level of exports to imports
- D/D^* : Logged country's real domestic absorption relative to the rest-of-the-world
- Y/Y^* : Logged real relative expenditure-side GDP at current PPPs relative to the rest-of-the-world

In the two-country analysis, rest-of-the-world output price level or domestic absorption is measured by the trade weighted average of non-US countries. Trade weights are from BIS and updated every 4 years during 1993-2016. For the period of 1970-1992, weights from 1993-1995 are used, while for the period of 2017-2019 weights from 2014-2016 are used.

B Empirical Robustness: Capital flows and trade

In this section we check the robustness of the findings in several dimensions. First, we redo the analysis using a different data source. Second, we consider aggregating the countries into a few country groups or including more countries. Third, we compare the patterns in real and nominal terms. Finally, we use the current account and alternative measures of the trade balance that down-weight small economies.

First, we evaluate PWT in comparison to the System of National Accounts from United Nation Statistics Division (UN SNA). To construct trade to output ratio (*TRY*), trade balance to output (*TBY*), and export-import ratio ($\ln X/M$) using UN SNA, we use GDP by expenditure at constant 2015 USD prices.

Comparing two different data sources, PWT and UN SNA, suggests potential concerns with using PWT data to study trade across countries over time, especially for entrepot and commodity exporting countries, as discussed in [Johnson et al. \(2013\)](#). For example, Norway's

TBY of 1970 when measured with PWT exceeds -50%, while it is around 0% with UN SNA. In our sample, there are seven countries with a differences in real TBY larger than 20% at least once during the sample period, potentially attributing to the measurement errors and terms of trade effects when constructing real terms. In Figure B.1, we plot the real TBY of these countries over time, measured with PWT (solid line) and UN SNA (dashed line). For the baseline analysis, we exclude these seven countries and only consider 36 countries in our baseline sample.

However, such errors are not crucial for the observed relationship with cross-sectional TBY dispersion with the level of trade. In Table B.1 we compare the elasticity using data with 43 countries when we take a year as an unit of observation, measured with PWT (columns 1 and 3) to the one measured with UN SNA (columns 2 and 4). The results from two sources are similar, suggesting that pattern we observe in the left panels of Figures 2 and 4 is robust to the source of data. On the other hand, when we take a country as an observation, these seven countries indeed are outliers, as shown in the lower panels of Figures 2 and 4.

When using a country as a unit of account, we find the positive between net and gross trade flows relationship holds (Figure 2) when we look at alternative windows. To show this we further split the sample by considering 15-year windows for each country (Figure B.2). Similar to the earlier findings, there is a positive relationship between the trade balance dispersion and level of trade, and a negative relationship between the dispersion of the export-import ratio and trade.

We also highlight several countries with very high levels of trade and very volatile trade balances. In terms of high levels of trade, we see that key entrepots like Belgium, Hong Kong, and Singapore stand out. In terms of highly volatile trade balance, it is Norway and Saudi Arabia in the periods from 1970-85 that stand out. Obviously these outliers arise from very substantial asymmetric shocks related to oil discoveries and the price of oil.

Second, we aggregate the countries into a few groups and examine the pattern in the aggregated data. We consider eight group of countries consisting of North America, Mexico, other South America, Europe, China, Japan, other Asia, and rest of the world. Using bilateral trade from DOTS, we construct the trade variables including TBY, export-import

ratio, and TRY for these groups. Figure B.3 shows that the relationship holds with the group of countries.

Our results on the comovement between trade integration and trade balance dispersion are robust to including more countries. Figure B.4 shows that dispersion in borrowing and lending is also rising when we consider a broader set of the 157 countries in the Penn World Table from 1950 onwards.

Third, we look at the sensitivity of the pattern measured in real terms. Figure B.5 shows the relationship we find using nominal terms also holds in real terms.

Finally, we confirm that when we use current account as a measure of net flows the positive correlation, we still find the positive correlation as we find in Figure 2. Figure B.6 shows the interquartile range of net trade flows over time, measured by the ratio of either trade balance or current account to GDP. Although there exists minor differences in these measures due to the differences between trade balance and current account – net income and net transfers – the two measures of dispersion move similarly over time, implying the positive relationship with the level of trade.

We also show the relationship between dispersion in net flows is a bit stronger if we measure the trade balance as a share of world average GDP (Figure B.7). This approach has the advantage of down-weighting small countries with large imbalances. In the upper panel of Figure B.7, we show the elasticity with a year as an observation, while the lower panel shows the results with a country as an observation. We again see the pattern is robust to the different measurement of trade. We find that, over the range of the changes in trade integration, the dispersion increases even more than in our main measure. Thus, our findings are robust to using an alternative weighting.

C Decomposition

Our analysis in section 3 shows that dispersion in trade balance has risen over time and that this rise occurs with an increase in trade and decline in dispersion in the trade balance as a share of trade (TBTR). Here we undertake a formal variance decomposition of the evolution of the variance of the trade balance as a share of GDP. This variance decomposition

introduces some additional terms but also makes clear how these terms are related to our two main variables: TRY and TBTR. We then run some reduced form regressions.

C.1 Variance Decomposition

Equation 1 allows us to decompose the variance of the trade balance to GDP ratio (TBY) into six terms:

$$\begin{aligned} \text{Var}(\text{TBY}) = & \underbrace{\mathbb{E}[\text{TRY}]^2 \cdot \text{Var}(\text{TBTR})}_{\text{Term 1}} + \underbrace{\text{Var}(\text{TRY}) \cdot \text{Var}(\text{TBTR})}_{\text{Term 2}} + \\ & \underbrace{\mathbb{E}[\text{TBTR}]^2 \cdot \text{Var}(\text{TRY})}_{\text{Term 3}} - \underbrace{2 \cdot \mathbb{E}[\text{TBTR}] \cdot \mathbb{E}[\text{TRY}] \cdot \text{Cov}(\text{TBTR}, \text{TRY})}_{\text{Term 4}} \quad (7) \\ & - \underbrace{\text{Cov}(\text{TBTR}, \text{TRY})^2}_{\text{Term 5}} + \underbrace{\text{Cov}(\text{TBTR}^2, \text{TRY}^2)}_{\text{Term 6}} \end{aligned}$$

The first term, $\mathbb{E}[\text{TRY}]^2 \text{Var}(\text{TBTR})$, captures the direct effect of trade on dispersion in the trade balance. Our decomposition of the inter-quartile range holding trade constant was meant to capture this effect. The second term shows that dispersion in trade influences dispersion in the trade balance. The next two terms depend on the mean trade balance, which should be equal to zero, although there can be some departures owing to measurement, sample composition, and the fact that we divide by each country's trade. Terms 5 and 6 are covariance and higher order terms.

Before using this decomposition, we study the changes in some specific terms over time in Figure C.1. First, as expected, the mean trade balance as a share of trade is very close to zero, $\mathbb{E}[\text{TBTR}] \approx 0$, while the squared mean of trade, $\mathbb{E}[\text{TRY}]^2$, and the variance of trade, $\text{Var}(\text{TRY})$, have almost tripled over time, and finally the variance of the trade balance as a share of trade, $\text{Var}(\text{TBTR})$, has fallen. Turning to the variance decomposition, Figure C.2 depicts the share of the variance accounted for by each term. It is evident that the main determinants of trade balance dispersion are terms 1 and 2, and that their stable contribution reflects a rising contribution from gross trade factors and declining contribution of net trade. Term 6 offsets some of the movements in Term 1 the 1980s but has fallen since the 1970s.

This variance decomposition makes clear that focusing on the mean of trade and the

standard deviation of the trade balance to trade ratio captures the first-order determinant of the variance of the trade balance to GDP ratio. It also shows that there are some additional terms that can move around over time but are not changing dramatically in the fifty year period that we study.

C.2 Reduced Form Analysis

We now show that trade integration remains the key explanatory variable even when we bring other variables on global asymmetries into the analysis. Specifically, we undertake a reduced-form regression analysis that relates the time-series variation in the cross-country dispersion in the trade balance to the time-series variation in trade integration and other business cycle variables. Given our variance decomposition from Equation 7 its clear that any reduced form decompositions are likely to be mis-specified. Nonetheless, we presents the results in Table C.1.

The share of trade to output, TRY, plays a crucial role in explaining the variation in dispersion in the trade balance, TBY, over time; it alone explains about 50 percent of the variation in the annual dispersion in the trade balance, as shown in the R-square of regression (1). Recall from the first term in Equation 7 that the coefficient on trade should be equal to the standard deviation of the standard deviation of the trade balance to trade if Terms 2-6 are zero. Including other regressors, such as median output growth, output growth dispersion, real exchange rate dispersion, and oil prices, as in regressions (2)-(7), raises the explanatory power only marginally. Furthermore, excluding the trade share as in the regression (8)-(9) lowers the R-squared significantly. In all cases, the mean trade-to-output ratio is significant, implying a percentage point higher TRY is related to 0.093-0.126 percentage point higher dispersion in TBY.

A competing hypothesis is that financial integration that allows for larger foreign holdings of domestic assets has enabled more borrowing and lending. To explore this mechanism, we introduce a measure of gross financial flows from Milesi-Ferretti (2021), measured by the median annual gross position as a share of GDP, $\frac{A_{it}+L_{it}}{Y_{it}}$, where A_{it} denotes gross foreign assets, and L_{it} denotes gross foreign liabilities. For the median country, these gross positions have grown substantially over time, from about 50 percent in 1970 to 300 percent in 2019.

In column (10), we see that this variable alone explains about 20% of the dynamics of the dispersion in borrowing and lending. However, when we include our trade measure too in column (11), we find trade to be more important, and financial integration now comes in negative, and the explanatory gain over including our trade measure alone is about nine percentage points.

D Robustness

D.1 Persistence of Productivity Shocks

In this section, we analyze the results of the model when we vary the persistence of productivity shocks. In particular, we increase the persistence of common productivity shocks from $\rho_{zc} = 0.96$ to 0.98. We adjust the volatility to keep the unconditional variance of the shock processes constant. The other parameters are kept the same as in the baseline. Table D.1 shows the results. The cases with higher persistence show almost same elasticity of dispersion to trade. For example, in both cases, the dispersion in the trade balance to output and the export-import ratio change by around 0.20 and -0.39 as gross trade increases by 1 percentage point. When the differential productivity is more persistent and countries are subject to larger idiosyncratic shocks, the dispersion in output across countries becomes stronger while the dispersion in domestic absorption gets smaller. The business cycle moments are slightly differ even though the shock process itself has the same variance. However, all of these changes are quantitatively small.

D.2 Number of Countries

Given that we are allowing the level of trade and trade costs to vary quite substantially we solve the model with a second order approximation. This level of approximation yields more accurate solutions than the linear approximation. However, a challenge with high-order approximations is that computational time increases quite substantially with the number of countries. Thus, for now we focus on estimating the effects by varying the number of countries in the model. We simulate the model with 10 countries given the parameters we

used in the baseline analysis. The findings show that the results are insensitive to adding the number of countries in the model.

To consider a larger set of countries, we let the two additional countries to be median sizes of the original distribution. The rest of the parameters are set as in the baseline. We take the same approach as in the baseline analysis to simulate and derive the model predictions. Figure D.4 shows the results and Table D.2 compares the moments with the baseline case with 8 countries. The result successfully captures the changes in the trade balance dispersion observed in data, growing by 0.08 as trade grows by 1 percentage points. Also, the relationship between TRY and X/M dispersion is negative.

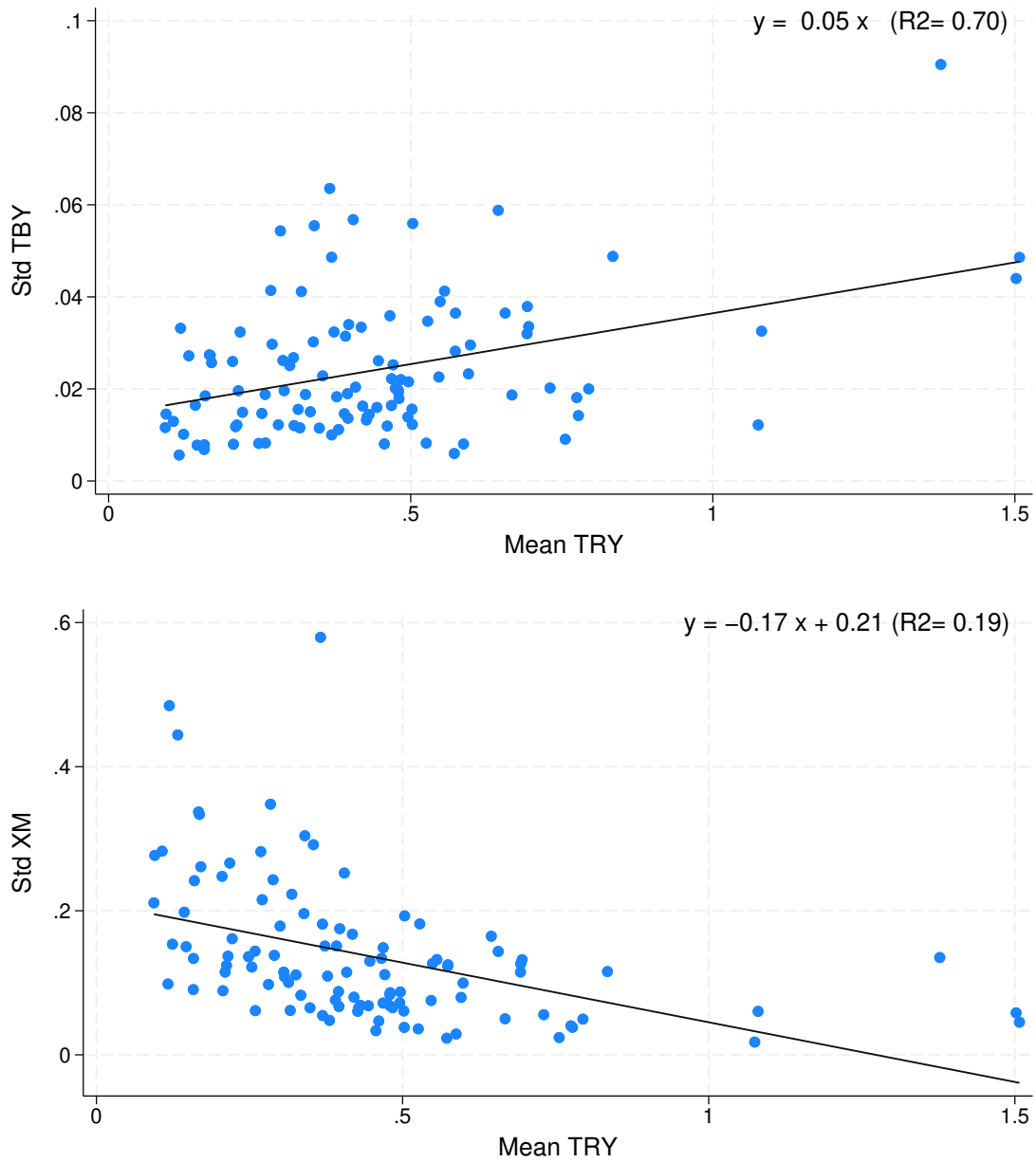
We then run the counterfactual exercises without financial or trade integration. The left panel of Table D.2 presents the results. When we shut off the financial shocks (green squares) the trade dispersion become is smaller and less elastic to gross trade. More importantly, without shocks to trade costs (red triangles) we cannot generate enough variations in gross trade, and there is an excess dispersion in net trade. The results show that the role of financial and trade integration for generating the increase in net and gross trade flows are robust to number of countries in the model.

APPENDIX - FIGURES AND TABLES

Figure B.1: Countries with Large Differences in Real TBY



Figure B.2: Relationship by Country Across Time Periods



Notes: Each observation is dispersion measured across countries in 1970-1984, 1985-1999, or 2000-2019.

Figure B.3: Relationship with Aggregated Group of Countries

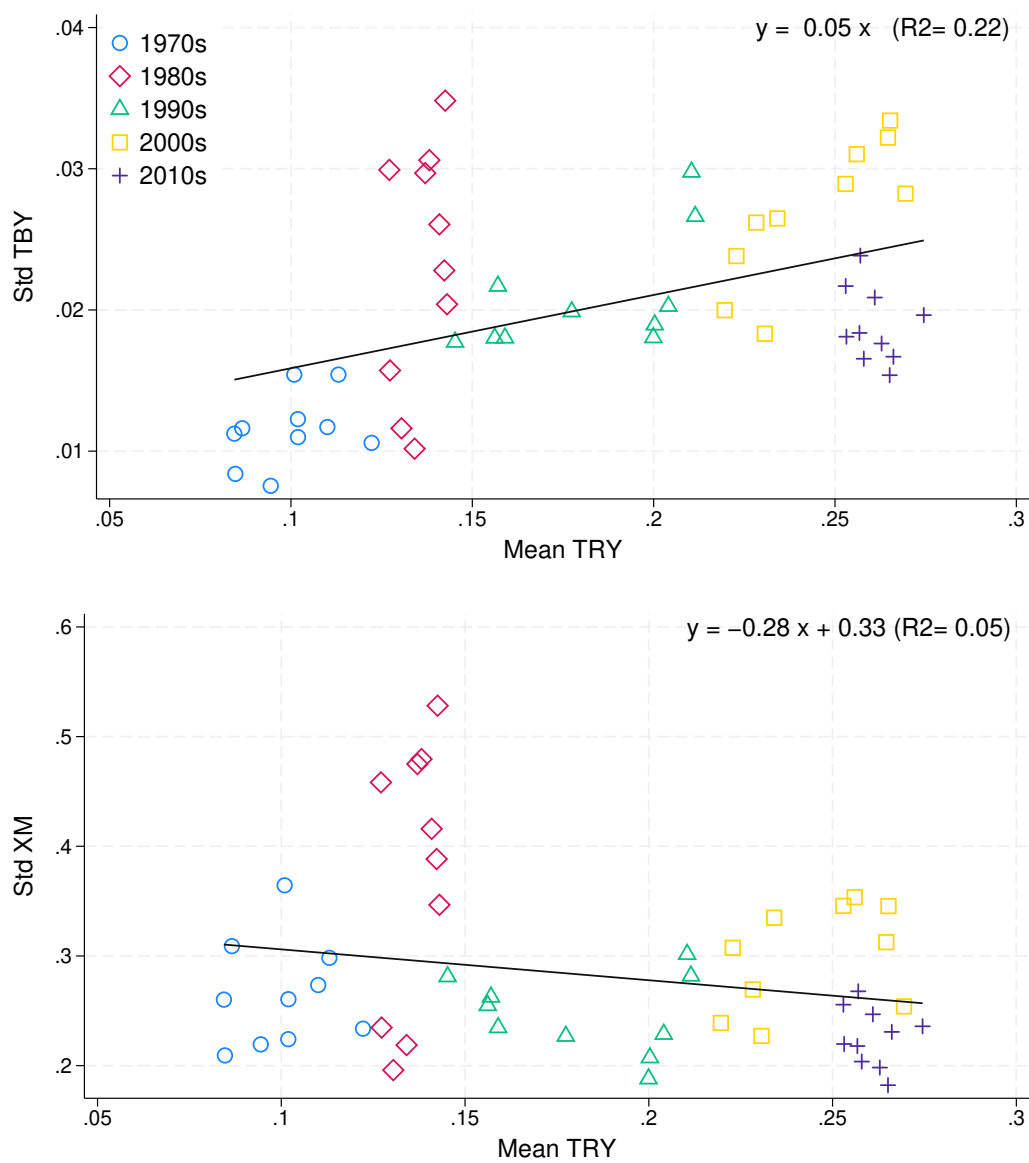
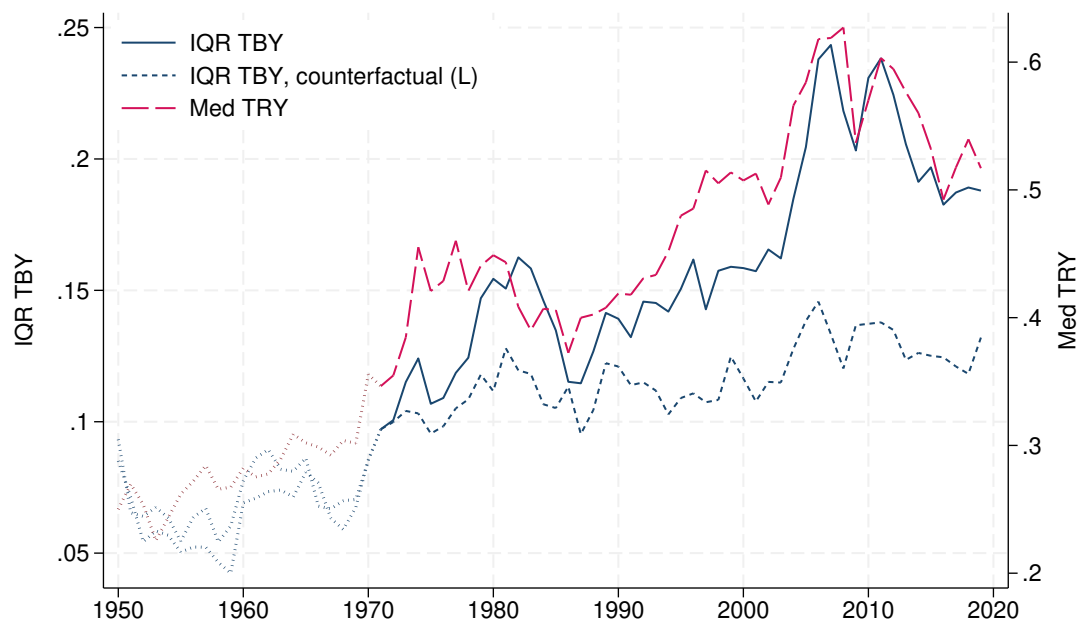
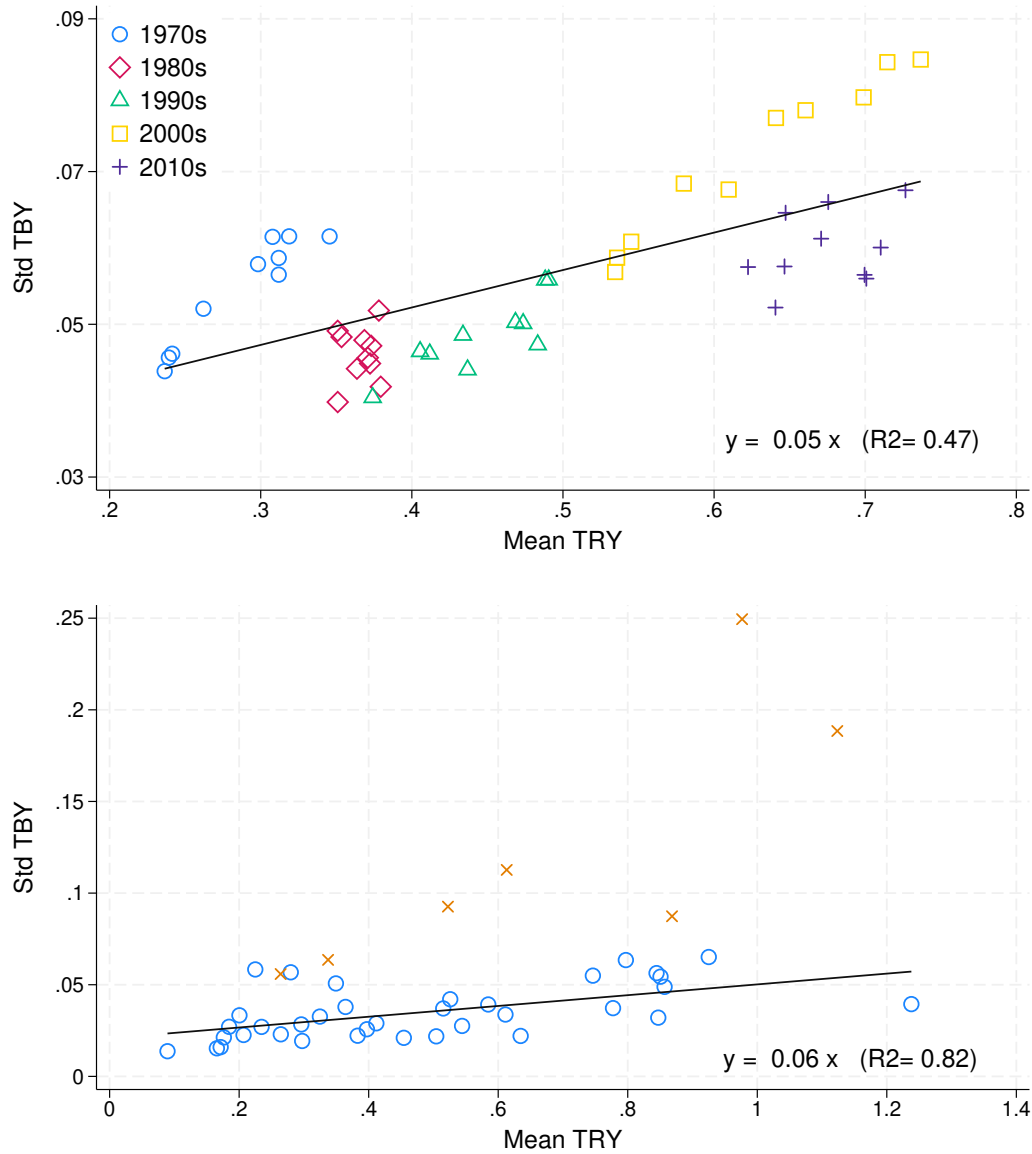


Figure B.4: Relationship in Larger Sample of Countries



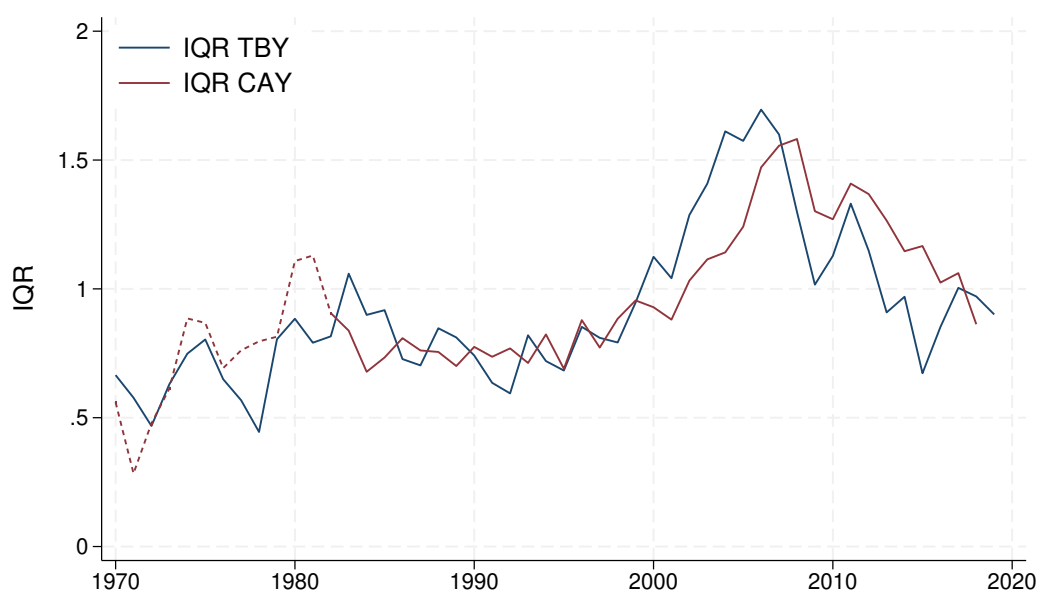
Notes: The figure shows the interquartile range of TBY and median of TRY for the full sample of 157 countries in PWT 10.0. Prior to 1970, there are countries with missing values resulting in an unbalanced panel and we plot them with dotted lines.

Figure B.5: Relationship in Real Terms



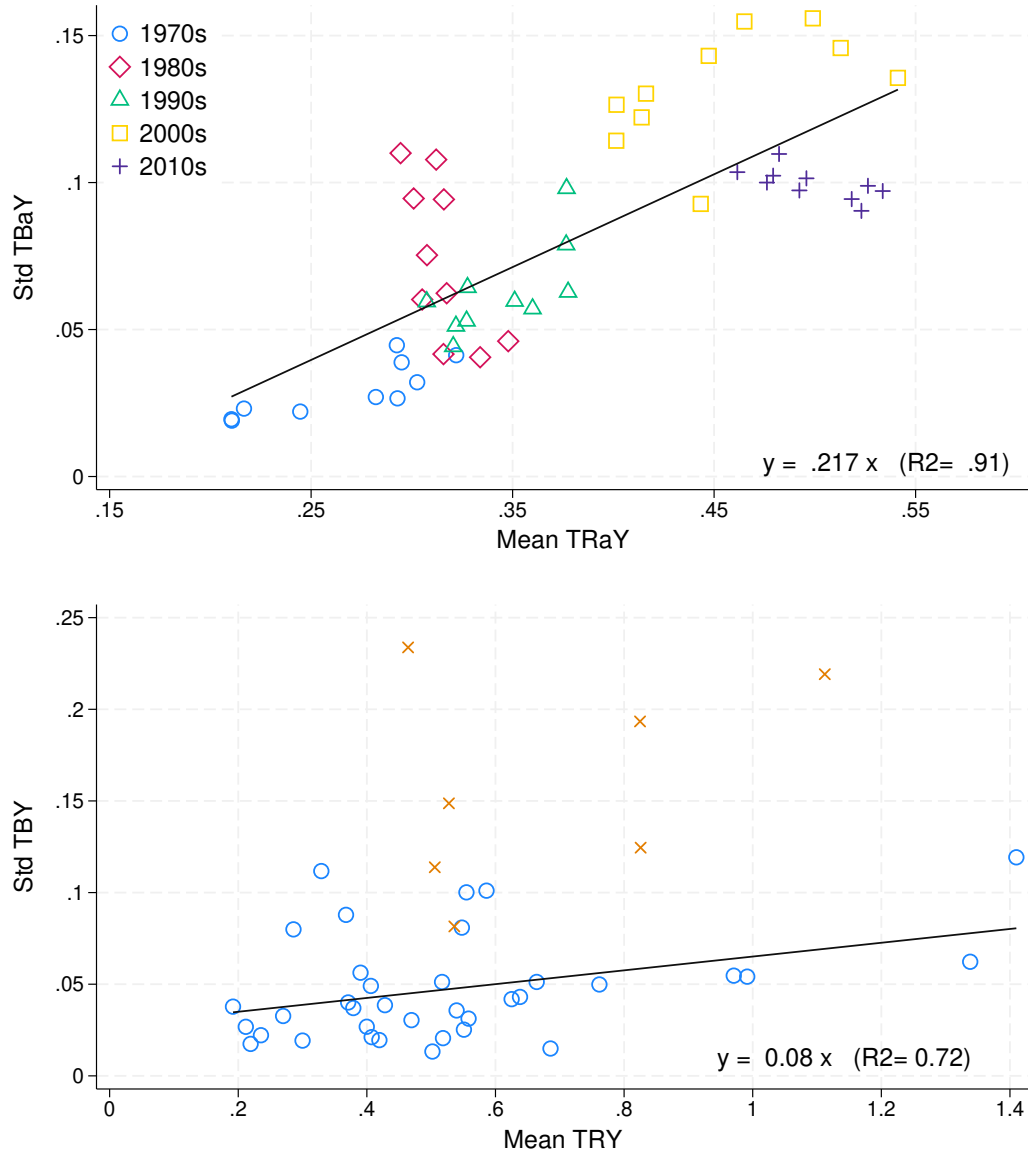
Notes: The upper panel plots cross-country moments for each year, with decades presented in different colors. The lower panel shows the time series moments for each country. The x-markers in the lower panel represent entrepot countries that are not included in our baseline sample and are not taken into account for the regression. Both TBV and TRY are measured in real terms.

Figure B.6: Relationship Measured by Current Account



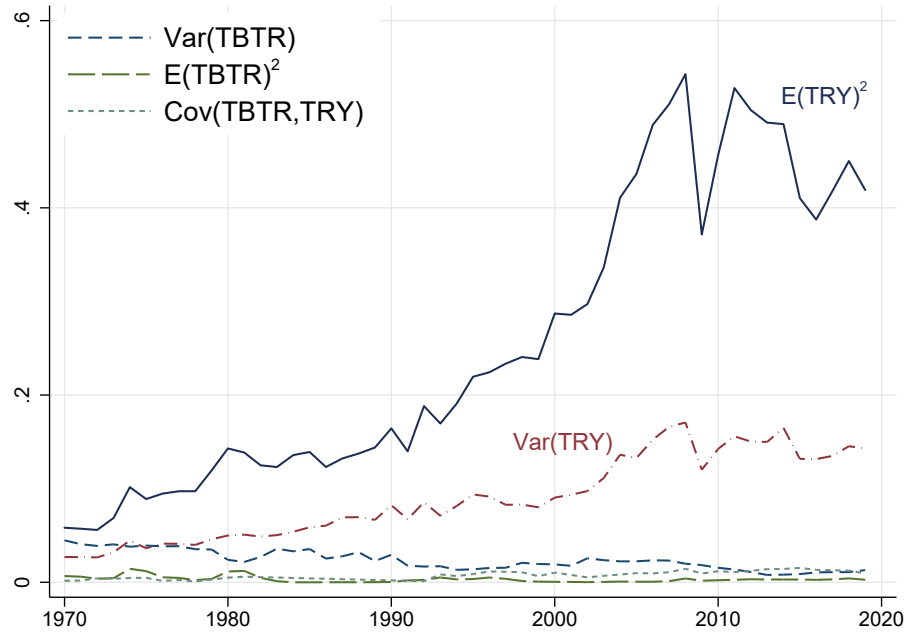
Notes: The figure shows the interquartile range of TBY and CAY, where CAY is current account as a share of GDP, calculated using our baseline sample of 36 countries. Current account is from WDI. Prior to 1983, some countries in the baseline sample are missing values for the current account.

Figure B.7: Relationship in Share of World Average Output



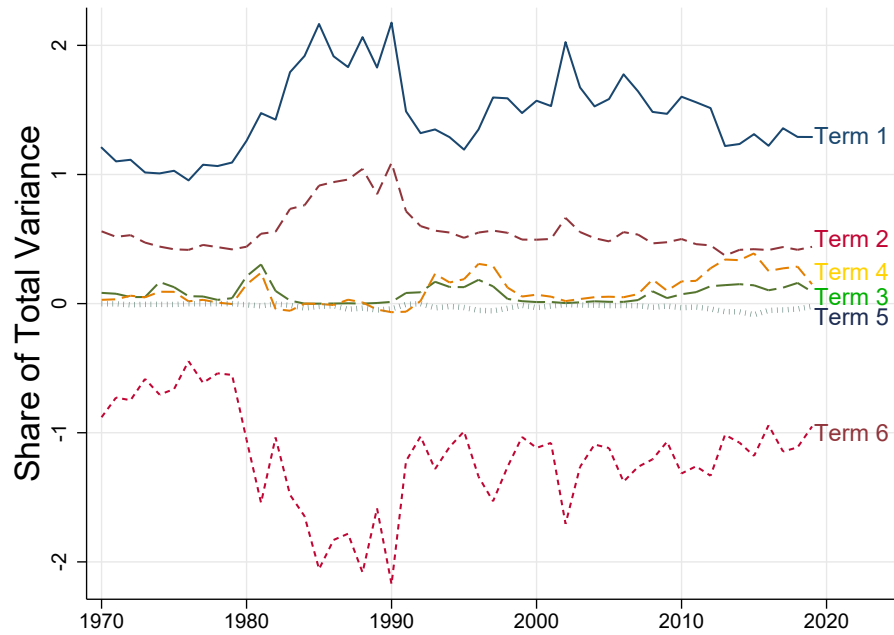
Notes: The figure shows the relationship between the net and gross flows as a share of world average output. In the upper panel the cross-country moments are calculated for each year, while in the lower panel takes each country as an observation and time series moments are calculated. The x-markers in the lower panel represent entrepot countries that are not included in our baseline sample and are not taken into account for the regression.

Figure C.1: Core Components in Variance Decomposition of TBY



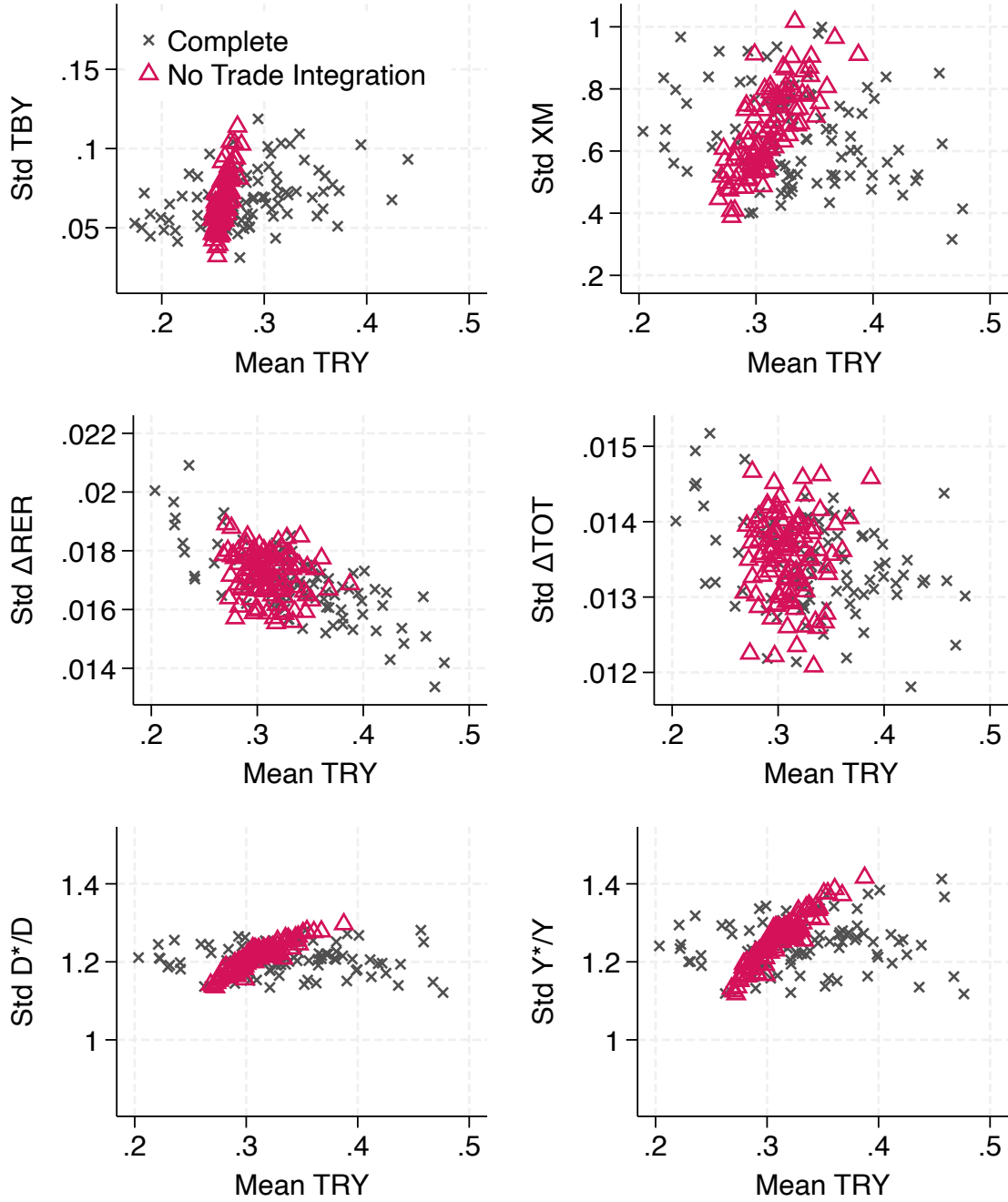
Notes: The figure shows the evolution of different moments of the distribution of gross and net trade over time.

Figure C.2: Contribution to Variance Decomposition of TBY



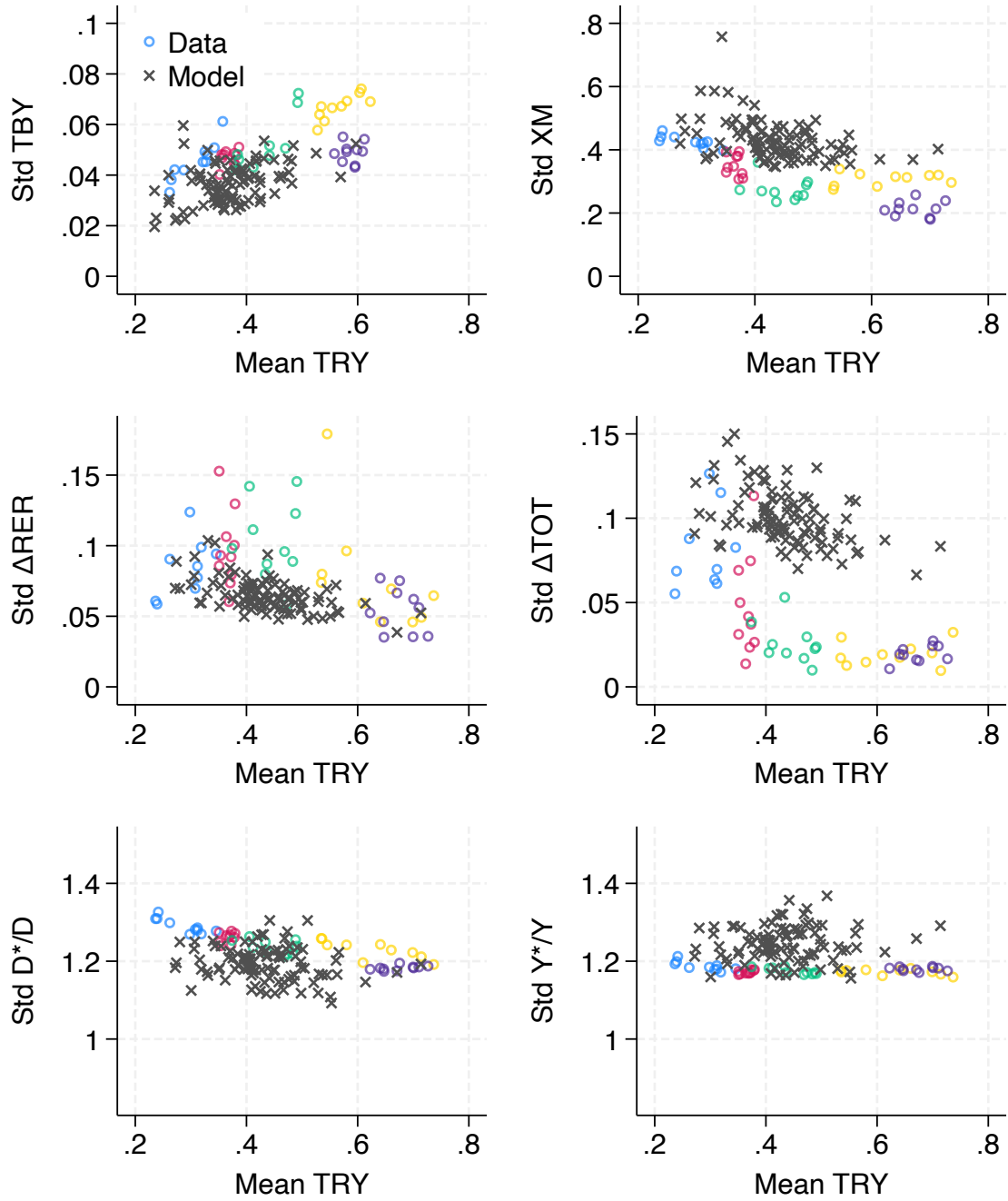
Notes: The figure shows the share of the variance of TBY accounted for each term in our variance decomposition from Equation 7.

Figure D.1: Sensitivity to Complete Market



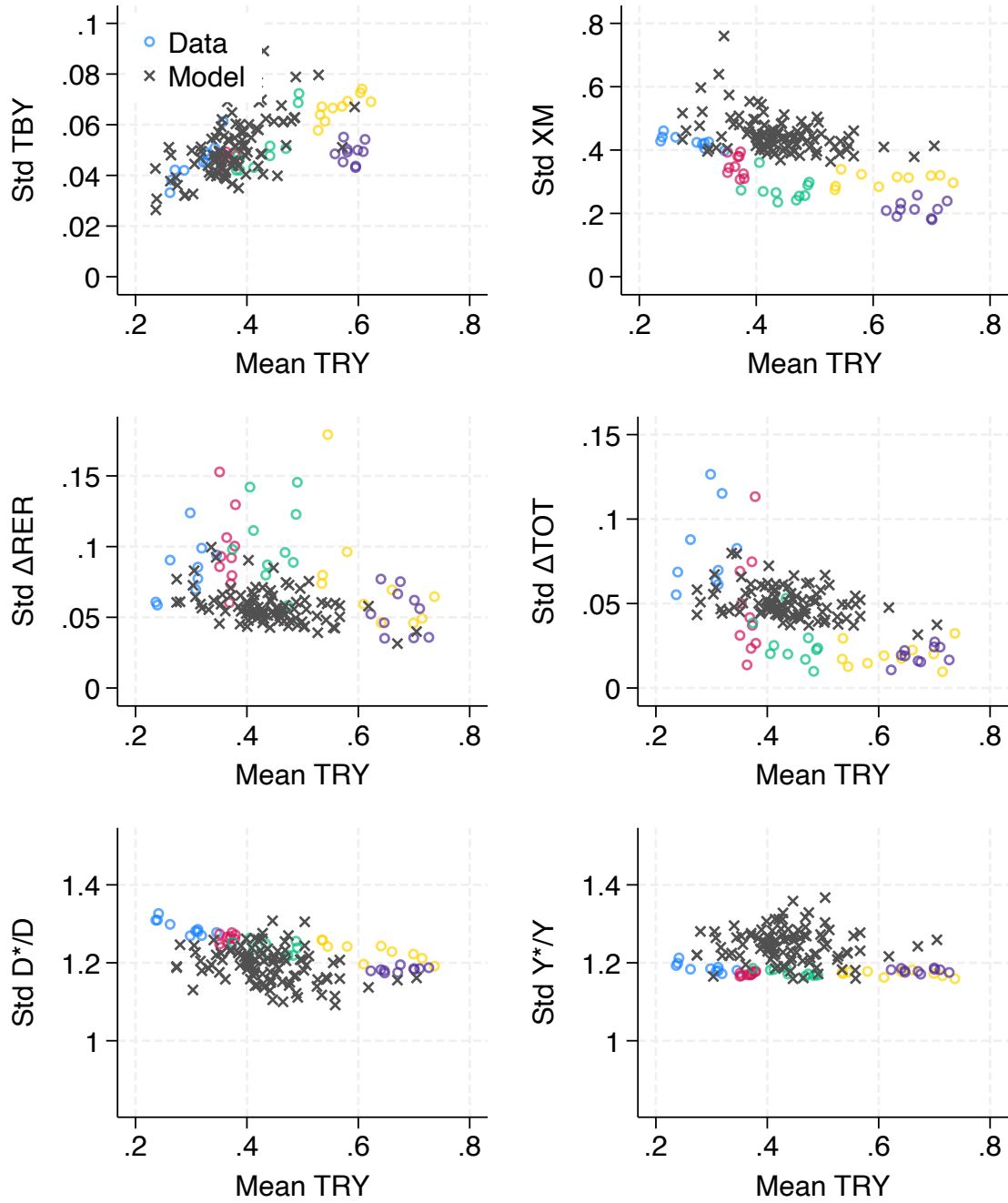
Notes: The graph shows the standard deviations net trade in relation to the mean of the gross trade. The colored dots show the data, the black crosses show the model prediction with the complete financial market, and the red triangles are those with the complete market but without trade integration, or $\sigma_{\xi_c} = 0$.

Figure D.2: Sensitivity to No Pricing-to-Market



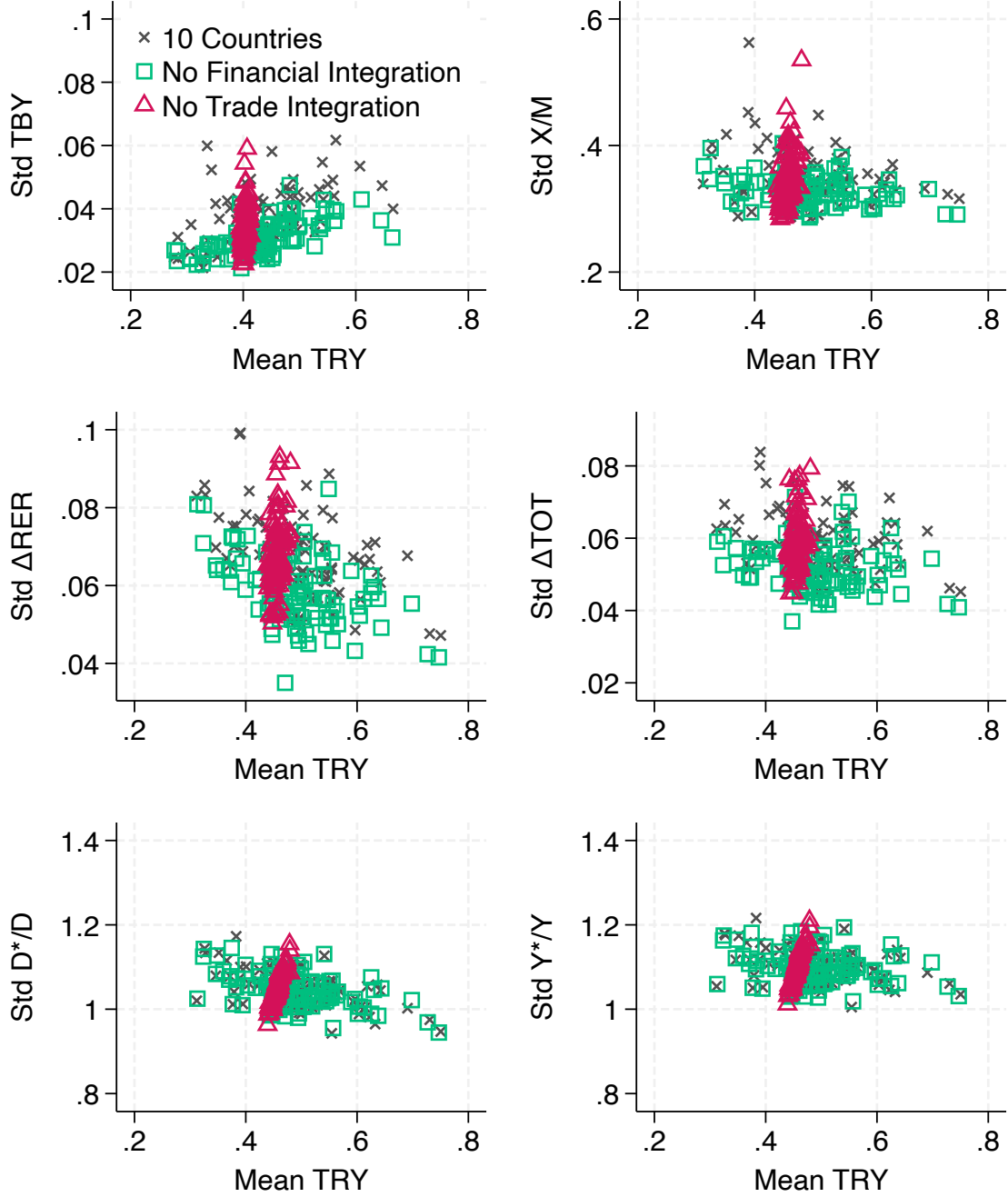
Notes: In each graph, y-axis is the standard deviation, and x-axis is trade share over output.

Figure D.3: Sensitivity to No Trade Adjustments



Notes: In each graph, y-axis is the standard deviation, and x-axis is trade share over output.

Figure D.4: Sensitivity to Number of Countries



Notes: The graph shows the standard deviations net trade in relation to the mean of the gross trade. The black crosses show the model prediction when we run the model 12 countries using our baseline parameters. The red triangle dots are those without trade integration, or $\sigma_{\xi_c} = 0$.

Table B.1: Comparison of Data Sources

	(1) Disp TBV	(2) Disp TBV	(3) Disp lnX/M	(4) Disp lnX/M
TRY mean	0.15*** (0.00)	0.10*** (0.00)	-0.12 (0.09)	-0.43*** (0.03)
Constant			0.41*** (0.04)	0.50*** (0.02)
Observations	50	50	50	50
R-squared	0.97	0.95	0.04	0.83
Data source	PWT	UN SNA	PWT	UN SNA

Notes: The table shows the regression result using 43 countries. Dispersion is measured by the standard deviation. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table C.1: Cross-sectional Regressions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
<i>Med</i> (<i>TRY</i>)	0.100*** (0.016)	0.101*** (0.016)	0.093*** (0.017)	0.104*** (0.016)	0.126*** (0.022)	0.097*** (0.016)	0.118*** (0.027)				0.212*** (0.041)
<i>Med</i> ($\Delta \ln Y$)		0.060 (0.066)				0.067 (0.061)	0.073 (0.067)	0.078 (0.083)	0.068 (0.084)		
<i>Med</i> ($\Delta \ln Y_{-1}$)		0.044 (0.070)				0.037 (0.067)	0.002 (0.081)	0.055 (0.082)	0.114 (0.081)		
$\ln$ Oil Price			0.005 (0.003)			0.006 (0.004)	0.004 (0.004)	0.018*** (0.006)	0.016*** (0.005)		
$\sigma(\Delta \ln Y)$				0.215 (0.166)	0.131 (0.166)	0.180 (0.144)	0.114 (0.153)	0.017 (0.172)	0.205 (0.171)		
$\sigma(\Delta \ln Y_{-1})$				0.074 (0.157)	0.101 (0.156)	0.008 (0.162)	0.049 (0.167)	-0.104 (0.211)	-0.142 (0.206)		
$\sigma(\Delta \ln RER)$					0.125* (0.062)		0.092 (0.077)		-0.202** (0.099)		
$\sigma(\Delta \ln RER_{-1})$					0.096 (0.077)		0.096 (0.092)		-0.155* (0.090)		
<i>Med</i> ((<i>A</i> + <i>L</i>)/ <i>Y</i>)										0.013*** (0.003)	-0.021*** (0.006)
Observations	48	48	48	48	48	48	48	48	48	48	48
R-squared	0.494	0.515	0.508	0.521	0.549	0.550	0.567	0.172	0.301	0.296	0.588

Notes: The dependent variable is the dispersion in trade balance to output ratio, *TBY*, measured by its interquartile range. Data is from Penn World Table 10.0, during 1970-2019 with 36 countries. *Med* denotes median, and σ denotes the interquartile range. Oil price is an annual average of imported crude oil price (\$/barrel, real) (US EIA). Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table D.1: Persistence of Productivity Shocks

		Data	Baseline	$\rho_{zc} = 0.98$
Integration and Cross-country Dispersion				
TBY	slope	0.11	0.09	0.09
X/M	slope	-0.39	-0.25	-0.25
X/M	constant	0.50	0.52	0.52
ΔRER	slope	-0.09	-0.08	-0.08
ΔRER	constant	0.13	0.11	0.11
ΔTOT	slope	-0.12	-0.05	-0.05
ΔTOT	constant	0.10	0.08	0.08
D/D^*	slope	-0.22	-0.12	-0.12
D/D^*	constant	1.33	1.24	1.24
Y/Y^*	slope	-0.01	0.03	0.03
Y/Y^*	constant	1.17	1.23	1.23
Business Cycle Moments				
$\sigma(\Delta y)$, %		3.69	3.58	3.56
$\sigma(\Delta c)/\sigma(\Delta y)$		1.06	0.78	0.79
$\sigma(\Delta inv)/\sigma(\Delta y)$		3.16	3.18	3.18
$\rho(\Delta y_t, \Delta y_{t-1})$		0.39	0.21	0.21
$\rho(\Delta c_t, \Delta c_{t-1})$		0.37	0.37	0.37
$\sigma(TRY)$		0.15	0.08	0.08
$\rho(TRY)$		0.98	0.70	0.70
$\rho(TBY, \Delta y)$		-0.01	0.00	0.00
$\rho(\Delta(c - c^*), \Delta RER)$		-0.21	-0.22	-0.22
$\rho(\Delta y, \Delta y^*)$		0.21	0.21	0.21
Trade Elasticity				
Short Run		0.22	1.54	1.54
Long Run		3.22	3.29	3.29

Notes: All of the moments except the slope for TBY are calculated using real values. Business cycle moments are averages across countries.

Table D.2: Model Results – Number of Countries

		Data	10 Countries		
			All	No financial integration	No trade integration
Integration and Cross-country Dispersion					
TBY	slope	0.11	0.08	0.07	0.09
X/M	slope	-0.39	-0.12	-0.08	2.23
X/M	constant	0.50	0.41	0.37	-0.67
ΔRER	slope	-0.09	-0.06	-0.05	0.35
ΔRER	constant	0.12	0.10	0.08	-0.09
ΔTOT	slope	-0.13	-0.03	-0.02	0.22
ΔTOT	constant	0.10	0.07	0.06	-0.04
D/D^*	slope	-0.22	-0.14	-0.11	4.10
D/D^*	constant	1.33	1.16	1.15	-0.77
Y/Y^*	slope	-0.01	-0.26	-0.24	3.72
Y/Y^*	constant	1.17	1.17	1.16	-0.65
Business Cycle Moments					
$\sigma(\Delta y)$, %		3.69	3.66	3.46	3.64
$\sigma(\Delta c)/\sigma(\Delta y)$		1.06	0.80	0.82	0.79
$\sigma(\Delta inv)/\sigma(\Delta y)$		3.16	3.40	3.06	3.29
$\rho(\Delta y_t, \Delta y_{t-1})$		0.39	0.20	0.23	0.20
$\rho(\Delta c_t, \Delta c_{t-1})$		0.37	0.36	0.37	0.36
$\sigma(TRY)$		0.15	0.08	0.08	0.01
$\rho(TRY)$		0.98	0.69	0.70	0.47
$\rho(TBY, \Delta y)$		-0.01	0.00	0.02	-0.01
$\rho(\Delta(c - c^*), \Delta RER)$		-0.21	-0.24	-0.22	-0.25
$\rho(\Delta y, \Delta y^*)$		0.21	0.20	0.23	0.20
Trade Elasticity					
Short Run		0.22	1.54	1.54	1.54
Long Run		3.22	3.31	3.34	3.31

Notes: All of the moments except the slope for TBY are calculated using real values. Business cycle moments are averages across countries.

Table D.3: Model Results – Complete Market and Counterfactuals

		Non-contingent bond			Complete market	
		Data	Baseline	No trade integration	All shocks	No trade integration
Integration and Cross-country Dispersion						
TBY	slope	0.11	0.09	0.09	0.24	0.25
X/M	slope	-0.39	-0.25	1.14	-0.59	4.46
X/M	constant	0.50	0.52	-0.05	0.85	-0.72
ΔRER	slope	-0.09	-0.08	0.04	-0.02	-0.01
ΔRER	constant	0.13	0.11	0.06	0.02	0.02
ΔTOT	slope	-0.12	-0.05	-0.03	0.00	0.00
ΔTOT	constant	0.10	0.08	0.08	0.01	0.01
D/D^*	slope	-0.22	0.03	4.51	0.11	2.50
D/D^*	constant	1.33	1.23	-0.56	1.21	0.48
Y/Y^*	slope	-0.01	-0.12	4.22	-0.04	1.37
Y/Y^*	constant	1.17	1.24	-0.49	1.22	0.79
Business Cycle Moments						
$\sigma(\Delta y)$, %		3.69	3.58	3.56	3.23	3.21
$\sigma(\Delta c)/\sigma(\Delta y)$		1.06	0.78	0.78	0.74	0.74
$\sigma(\Delta inv)/\sigma(\Delta y)$		3.16	3.18	3.08	1.70	1.67
$\rho(\Delta y_t, \Delta y_{t-1})$		0.39	0.21	0.22	0.33	0.33
$\rho(\Delta c_t, \Delta c_{t-1})$		0.37	0.37	0.37	0.58	0.57
$\sigma(TRY)$		0.15	0.08	0.01	0.06	0.02
$\rho(TRY)$		0.98	0.70	0.37	0.58	0.39
$\rho(TBY, \Delta y)$		-0.01	0.00	-0.01	-0.04	-0.04
$\rho(\Delta(c - c^*), \Delta RER)$		-0.21	-0.22	-0.22	1.00	1.00
$\rho(\Delta y, \Delta y^*)$		0.21	0.21	0.22	0.33	0.33
Trade Elasticity						
Short Run		0.22	1.54	1.54	1.57	1.58
Long Run		3.22	3.29	3.29	3.24	3.24

Notes: All of the moments except the slope for TBY are calculated using real values. Business cycle moments are averages across countries.